Cardinal Characteristics

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I will divide this document into three parts. The first chapter will explain the general theory of cardinal characteristics that we will use, the second chapter the rearrangement number, and the third chapter will cover the density cardinal.

Summary

1 Cardinal Characteristics of the Continuum

1.1 Growth of Functions

Definição de $\mathfrak{b}$ e $\mathfrak{d}$ e demonstração de que

$$\aleph_1 \leq \mathrm{cf}(\mathfrak{b}) = \mathfrak{b} \leq \mathrm{cf}(\mathfrak{d}) \leq \mathfrak{d} \leq \mathfrak{c}$$

(Andreas Blass - Handbook of Set Theory [6])

1.2 Galois-Tukey Connections

Lets introduce the language of Galois Tukey Connections (GT).

Definition 1.1. A relational system is a triple $\mathcal{A} = (A_-, A_+, A)$, with $A \subseteq A_- \times A_+$ where, for every $a_- \in A_-$, there is $a_+ \in A_+$ such that $a_- A a_+$, and, for every $a_+ \in A_+$ there is $a_- \in A_-$ such that $a_- A a_+$ fails.

You can interpret A_- as the set of challenges you need to surpass, and A_+ as the responses you will use to pass the challenges. So, the relation A indicates when a challenge is passed. The other conditions requires that any challenge can be surpassed (this guarantees $\mathfrak{d}(\mathcal{A})$ is well defined), and that we don't have any response for every challenge, to avoid trivializations (and to guarantee $\mathfrak{b}(\mathcal{A})$ is well defined).

Any relational system $\mathcal{A} = (A_-, A_+, A)$ has a dual $A^\top = (A_+, A_-, \neg\check{A})$, where $a_+ \neg\check{A} a_-$ if $\neg(a_- A a_+)$.

Definition 1.2. Given a relation system $\mathcal{A} = (A_-, A_+, A)$, we associate the following cardinals

- the dominating number $\mathfrak{d}(\mathcal{A})$ is the smallest size of a family $C_+ \subseteq A_+$ s.t. for all $a_- \in A_-$, there is a $a_+ \in C_+$ with $a_- A a_+$;
- the unbounding number $\mathfrak{b}(\mathcal{A})$ is the smallest size of a family $C_- \subseteq A_-$ s.t. for all $a_+ \in A_+$, there is $a_- \in C_-$ s.t. $\neg(a_- A a_+)$.

Intuitively, the dominating number is the smallest size of a set of responses that do the job, and the unbounding number the smallest size of a set of challenges that "breaks" any given response. Note that $\mathfrak{b}(\mathcal{A})$ and $\mathfrak{d}(\mathcal{A})$ are dual in the sense that $\mathfrak{b}(A^\top) = \mathfrak{d}(A)$ and $\mathfrak{d}(A^\top) = \mathfrak{b}(A)$.

Sketch. Now, we want to define a reduction $\varphi : \mathcal{A} \to \mathcal{B}$ between two relational systems, which helps to prove that $\mathfrak{d}(\mathcal{A}) \leq \mathfrak{d}(\mathcal{B})$ and $\mathfrak{b}(\mathcal{A}) \geq \mathfrak{d}(\mathcal{B})$. So, it's natural to think like this: Since we want to reduce $\mathcal{A}$ to $\mathcal{B}$, we can work really well in $\mathcal{B}$, so, given a challenge $a_- \in A_-$, we want to convert it to a challenge $\varphi_-(a_-) \in B_-$, where we know it has an answer $b_+ \in B_+$ that responds this challenge, i.e., $\varphi(a_-) B b_+$, so it only remains to convert this response $b_+ \in B_+$ back to a response $\varphi_+(b_+) \in A_+$, where $a_- A \varphi_+(b_+)$.

So we can define:

Definition 1.3. Given two relational systems $\mathcal{A} = (A_-, A_+, A)$ and $\mathcal{B} = (B_-, B_+, B)$, $\mathcal{A}$ is Tukey reducible to $\mathcal{B}$ ($\mathcal{A} \leq_T \mathcal{B}$) if there are functions $\varphi_- : A_- \to B_-$, $\varphi_+ : B_+ \to A_+$ such that $\varphi_-(a_-) B b_+ \Rightarrow a_- A \varphi(b_+)$ for all $a_- \in A_-$ and $b_+ \in B_+$.

Pendente

1.3 Null and Meager Sets

The main result of this chapter is known as the Cichoń's Diagram (Figure 1)

$$\begin{tikzcd}
 & \mathrm{cov}(\mathcal{N}) \arrow[r, "(3)"] & \mathrm{non}(\mathcal{M}) \arrow[r, "(2)"] & \mathrm{cof}(\mathcal{M}) \arrow[r, "(?)"] & \mathrm{cof}(\mathcal{N}) \arrow[r, "(2)"] & 2^{\aleph_0} \\
 & & \mathfrak{b} \arrow[u, "(?)"] \arrow[r, "(1)"] & \mathfrak{d} \arrow[u, "(?)"'] & & \\
\aleph_0 \arrow[r, "(2)"'] & \mathrm{add}(\mathcal{N}) \arrow[uu, "(2)"] \arrow[r, "(?)"'] & \mathrm{add}(\mathcal{M}) \arrow[u, "(?)"] \arrow[r, "(2)"'] & \mathrm{cov}(\mathcal{M}) \arrow[u, "(?)"'] \arrow[r, "(3)"'] & \mathrm{non}(\mathcal{N}) \arrow[uu, "(2)"'] &
\end{tikzcd}$$

Fig. 1: Cichoń's Diagram (w/ clickable hyperlinks)

Definição de um ideal

Theorem 1.1. If $\mathcal{I}$ is a σ-complete ideal containing all singletons, then

$$\aleph_0 \leq \mathrm{add}(\mathcal{I}) \leq \mathrm{cov}(\mathcal{I}), \mathrm{non}(\mathcal{I}) \leq \mathrm{cof}(\mathcal{I}) \leq 2^{\aleph_0}$$

summarized by (Figure 2)

$$\begin{tikzcd}
 & & \mathrm{cov}(\mathcal{I}) \arrow[dr] & & \\
\aleph_0 \arrow[r] & \mathrm{add}(\mathcal{I}) \arrow[ur] \arrow[dr] & & \mathrm{cof}(\mathcal{I}) \arrow[r] & 2^{\aleph_0} \\
 & & \mathrm{non}(\mathcal{I}) \arrow[ur] & &
\end{tikzcd}$$

Fig. 2: Theorem 1.1

Definição de $\mathcal{M}$ e $\mathcal{N}$

Theorem 1.2. $\mathcal{M}$ and $\mathcal{N}$ are both ideals, in particular, Theorem 1.1 applies to them

Proof. Pendente $\dashv$

The following classical lemma shows that null and meager describe "smallness" in two different ways.

Lemma 1.1. There exists sets $A \in \mathcal{N}$ and $B \in \mathcal{M}$ such that $A \cup B = \mathbb{R}$.

Proof. Pendente $\dashv$

This easy lemma has two interesting consequences.

Theorem 1.3. (Rothberger - 1938)

$$\mathrm{cov}(\mathcal{M}) \leq \mathrm{non}(\mathcal{N}) \quad \text{and} \quad \mathrm{cov}(\mathcal{N}) \leq \mathrm{non}(\mathcal{M})$$

Proof. Pendente ⊣

Theorem 1.4. (Erdös-Sierpiński). If $\mathrm{add}(\mathcal{N}) = \mathrm{cof}(\mathcal{N})$, then, there exists a bijection $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(X) \in \mathcal{N} \iff X \in \mathcal{M} \quad \text{and} \quad f(X) \in \mathcal{M} \iff X \in \mathcal{N}$$

In particular, CH implies the existence of such bijection.

Proof. Pendente ⊣

2 The Rearrangement Number

In contrast to the growth functions related cardinal characteristics this one is really easy to motivate, even for undergrads. It's known by the Riemann Rearrangement Theorem that any conditionally convergent (c.c.) series can be rearranged so it converges to any real, diverges by $\pm\infty$ or oscillates. Another way to state this theorem is as a property of $\mathrm{Sym}(\omega)$:

Theorem 2.1. (Riemann Rearrangement) Given any c.c. series $\sum_n a_n$ and $\alpha, \beta \in \mathbb{R} \cup \{\pm\infty\}$, there is some permutation $p \in \mathrm{Sym}(\omega)$ such that

$$\limsup_m \sum_{n=0}^{m} a_{p(n)} = \alpha \quad \text{and} \quad \liminf_m \sum_{n=0}^{m} a_{p(n)} = \beta$$

One natural generalization one can think about is: how much smaller could we choose $\mathrm{Sym}(\omega)$? It turns out that we can define this in terms of cardinality as we will see, but it's far from trivial what this cardinal should be. We will prove this is a cardinal characteristic of the continuum, but it is still open if we can prove the equality with any well-known cardinal characteristic.

2.1 Definitions

Definition 2.1. $\mathfrak{rr}$ is the smallest cardinality of any family $C \subseteq \mathrm{Sym}(\omega)$ such that, for every c.c. series $\sum_n a_n$ of reals, there is some $p \in C$ for which $\sum_n a_{p(n)}$ no longer converges to the same limit.

Analogously, if $\sum_n a_{p(n)}$ is required to converge, diverge to $\pm\infty$ or diverge by oscillation, then we have new cardinals $\mathfrak{rr}_f$, $\mathfrak{rr}_i$ and $\mathfrak{rr}_o$, respectively.

We could also consider joints of those conditions, like $\mathfrak{rr}_{fi}$ where $\sum_n a_{p(n)}$ either converges or diverges to $\pm\infty$.

Its easy to see that, if $C \subseteq \mathrm{Sym}(\omega)$ satisfies, for example, $\mathfrak{rr}_i$, then it also satisfies $\mathfrak{rr}_{fi}$, so $\mathfrak{rr}_{fi} \leq \mathfrak{rr}_i$, in general, we can make the Hasse Diagram in Figure 3.

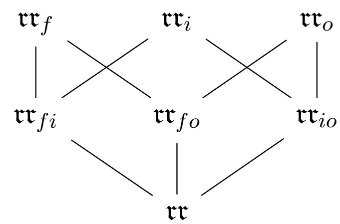

Fig. 3: Hasse Diagram of the Rearrangement Numbers

using this notation, $\mathfrak{rr}$ would be $\mathfrak{rr}_{fio}$, as expected.

2.2 Padding with Zeros

By Riemann's rearrangement $\mathfrak{rr} \leq |\mathrm{Sym}(\omega)| = \mathfrak{c}$, so, if $\mathfrak{rr}$ is uncountable, then its a cardinal characteristic of the continuum.

Indeed, to prove that $|\mathrm{Sym}(\omega)| = \mathfrak{c}$, note that $\mathrm{Sym}(\omega) \subseteq \omega^\omega$, therefore $|\mathrm{Sym}(\omega)| \leq \aleph_0^{\aleph_0} = 2^{\aleph_0}$, so we will construct an injection $\varphi : [0,1] \to \mathrm{Sym}(\omega)$, s.t. for each real number $a = (0.d_0 d_1 d_2 \dots)_2 \in [0,1]$ we associate $\varphi(a)$ as the bijection in ω that swaps $2n$ with $2n+1$ if $d_n = 1$, and does not swap them if $d_n = 0$.

In this subsection we will prove its actually a cardinal characteristic and we will also explain an important technique about padding zeros.

Theorem 2.2. $\mathfrak{rr}$ is uncountable.

Sketch. Let $C = \{p_n : n \in \omega\}$ be any countable subset of $\mathrm{Sym}(\omega)$ we will find a c.c. sequence that can't be corrupted by any of the permutations in C. To do this, we will start with any c.c. series, let's say $b_n = \frac{(-1)^n}{n}$ and we will pad a lot of zeroes between the terms and create a new sequence a_n such that $\sum_n a_n = \sum_n a_{p_m(n)}$, $\forall m \in \omega$. We will do this defining a new function $\ell(n)$ such that $a_n = 0$ if $n \notin \mathrm{ran}(\ell)$, so, if in the summation we take only the non zero terms, we will have to prove that

$$\sum_k a_{\ell(k)} = \sum_k a_{p_m(k)}$$

then, to force $\ell(1), \ell(2), \dots$ to eventually appear in the same order as the $p_m(k)$'s, we need to preserve the relative order of the terms $\ell(n)$ for almost all n, for example

$$p_m^{-1}(\ell(m)) < p_m^{-1}(\ell(m+1)) < p_m^{-1}(\ell(m+2)) < \dots$$

so, we want to avoid that $p_m^{-1}(\ell(k)) > p_m^{-1}(\ell(k+1))$, for some $m \leq k$. Then, to reach a contradiction, we could say that $\ell(k+1) \neq p_m(j)$, for any $p_m^{-1}(\ell(k)) > j$ and $m \leq k$.

Proof. Assume by contradiction that there is a $C = \{p_n : n \in \omega\} \subseteq \mathrm{Sym}(\omega)$, Let $b_n := \frac{(-1)^n}{n}$ (indeed, we can take any term of a c.c. series). Lets define $\ell : \omega \to \omega$ a fast growing function that will indicate the new position of b_k and will add a lot of 0's between the b_k's, i.e.

$$a_n := \begin{cases} b_k,\ n = \ell(k) \\ 0, \text{ otherwise} \end{cases}$$

So, define $\ell(0) = 0$ and, given $\ell(k)$ defined, let $\ell(k+1) > \ell(k)$ such that $\ell(k+1) \neq p_m(j)$, $j \leq p_m^{-1}(\ell(k))$, $m \leq k$ (which exists, since there are finite terms $p_m(j)$ of that form).

Now, we claim that $\forall m \geq k$, $p_m^{-1}(\ell(k)) < p_m^{-1}(\ell(k+1))$, in fact, if there was a $m \leq k$ s.t. $p_m^{-1}(\ell(k)) > p_m^{-1}(\ell(k+1))$, then, for $j = p_m^{-1}(\ell(k+1))$, by hypothesis

$$\ell(k+1) \neq p_m\big(p_m^{-1}(\ell(k+1))\big) = \ell(k+1)$$

a contradiction. So, we finally have that, $\forall m \in \omega$

$$p_m^{-1}(\ell(m)) < p_m^{-1}(\ell(m+1)) < p_m^{-1}(\ell(m+2)) < \dots$$

so, p_m^{-1} preserves the relative order of b_k's, except for possibly the first m terms, so, its clear that

$$\sum_n a_n = \sum_n a_{p_m(n)},\ \forall m \in \omega$$

⊣

2.3 Mixing Permutations

In this subsection we will prove some equalities on the variants of the rearrangement number by explaining an important technique called The Mix Lemma.

Lemma 2.1. (The Mix Lemma) For all $p, q \in \mathrm{Sym}(\omega)$, there is a $g \in \mathrm{Sym}(\omega)$ such that:

- $g[n] = p[n]$ for infinitely many n's;
- $g[n] = q[n]$ for infinitely many n's

To prove The Mix Lemma we will first prove the following lemma

Lemma 2.2. For any $p \in \mathrm{Sym}(\omega)$, if $h : n \to \omega$ is injective, then there exists $h \subseteq h' : M \to \omega$, with $M > n$, such that $\mathrm{Im}(h') = p[M]$.

Proof. Since p is a bijection, there are $m_0, m_1, \dots, m_{n-1}$ such that $p(m_i) = h(i)$, so, if we choose $M = \max\{m_0, m_1, \dots, m_{n-1}\} + 2 > n$, then $h[n] = \mathrm{Im}(h) \subseteq p[M]$ and $h' = p|_M$ do the job. ⊣

Now, let's prove the mix lemma

Proof. **(The Mix Lemma.)** Let $g_0 = \emptyset$. If $g_{2k-1} : n_{2k-1} \to \omega$ was already defined, by the previous lemma there exists a $g_{2k} : n_{2k} \to \omega$ extending g_{2k-1} s.t. $\mathrm{Im}(g_{2k}) = p[n_{2k}]$. Analogously, if g_{2k} was already defined, let $g_{2k+1} : n_{2k+1} \to \omega$ extending g_{2k} s.t. $\mathrm{Im}(g_{2k+1}) = q[n_{2k+1}]$.

Since $\{g_n : n \in \omega\}$ is a family of compatible functions, then $g := \bigcup_{n\in\omega} g_n$ is well defined. To show $g \in \mathrm{Sym}(\omega)$, note first that, since by the previous lemma $M > n$, then $n_k > n_m$, $\forall k > m$, therefore $\mathrm{dom}(g) = \omega$ and, since each g_k is injective, g is also. Now, for any $k \in \omega$, there is a $k' \in \omega$ s.t. $p(k') = k$ and, since g agrees with p for infinitely many n's, then k is eventually in the image of g. Therefore we have that $g \in \mathrm{Sym}(\omega)$. $\dashv$

Now, lets narrow down how much rearrangement numbers are there

Theorem 2.3.

$$\mathfrak{rr} = \mathfrak{rr}_{fo} = \mathfrak{rr}_{io} = \mathfrak{rr}_o$$

Proof. Since $\mathfrak{rr}_o \geq \mathfrak{rr}_{fo}, \mathfrak{rr}_{io} \geq \mathfrak{rr}$, its sufficient to show that $\mathfrak{rr} \geq \mathfrak{rr}_o$.

Let $C \subseteq \mathrm{Sym}(\omega)$ be as in the definition of $\mathfrak{rr}$, by The Mix Lemma 2.1, for each $p \in C$ there is a $g_p \in \mathrm{Sym}(\omega)$ s.t. g_p agrees with p and $q = \mathrm{id}$ for infinitely many n's. Define

$$C' := C \cup \{g_p : p \in C\}$$

By the previous theorem, $|C| > \aleph_0$, so $|C'| = |C| = \mathfrak{rr}$. We will show that C' satisfies $\mathfrak{rr}_o$.

Let $\sum_n a_n$ be c.c., we know that there exists a $p \in C$ that corrupts the series. If $\sum_n a_{p(n)}$ diverges by oscillation, we are done, otherwise, if it diverges for infinity, or converges to a different number, then $\sum_n a_{g_p(n)}$ has a subsequence going to infinity or this new sum, and another going to the original sum, so it diverges by oscillation. $\dashv$

The previous theorem narrows down the number of rearrangement cardinals (Figure 4)

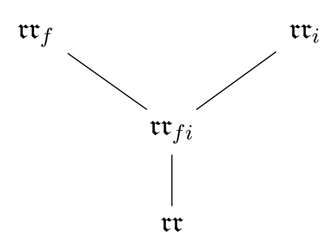

Fig. 4: Narrowed Hasse Diagram

2.4 Measure and Random Signs

In this subsection we relate the rearrangement number with the covering number for measure. In order to relate them we will need a consequence of Kolmogorov's Two Series Theorem, whose proof depends on the following inequality also due to Kolmogorov:

Lemma 2.3. (Kolmogorov's Inequality) Let $X_1, X_2, \dots, X_n$ be independent random variables with $\mathbb{E}[X_k] = 0$ and $\text{var}(X_k) < \infty$ for $k = 1, \dots, n$. Then, $\forall \varepsilon > 0$ we have that:

$$\mathbb{P}\left(\max_{1 \le k \le n} |S_k| \ge \varepsilon\right) \le \frac{\text{var}(S_n)}{\varepsilon^2} = \frac{1}{\varepsilon^2} \sum_{k=1}^{n} \text{var}(X_k)$$

where $S_k = X_1 + X_2 + \cdots + X_k$.

Proof. Let $E = \{\max_{1 \le k \le n} |S_k| \ge \varepsilon\}$ and consider the events

$$E_k = \{|S_1| < \varepsilon, |S_2| < \varepsilon, \dots, |S_{k-1}| < \varepsilon, |S_k| \ge \varepsilon\}$$

Clearly E_k are disjoint and $E = \bigcup_{k=1}^{n} E_k$, therefore

$$\mathbb{P}(E) = \sum_{k=1}^{n} \mathbb{P}(E_k) \tag{1}$$

Furthermore, since $\mathbb{E}[X_k] = 0$, $k = 1 \dots, n$, then $\mathbb{E}[S_n] = 0$. And, since S_n^2 is always greater or equal than 0, then $\mathbb{E}[S_n^2] \ge \mathbb{E}[S_n^2 \cdot I_A]$ for any event A, consequently:

$$\text{var}(S_n) = \mathbb{E}[S_n^2] \ge \mathbb{E}[S_n^2 \cdot \mathbf{1}_E] = \sum_{k=1}^{n} \mathbb{E}[S_n^2 \cdot \mathbf{1}_{E_k}] \tag{2}$$

Now, note that

$$S_n^2 = (S_n + (S_k - S_n))^2 = S_k^2 + 2S_k(S_n - S_k) + (S_n - S_k)^2$$

therefore, multiplying by $\mathbf{1}_{E_k}$ and taking the expected value at both sides

$$\mathbb{E}[S_n^2 \cdot \mathbf{1}_{E_k}] = \mathbb{E}[S_k^2 \cdot \mathbf{1}_{E_k}] + 2\mathbb{E}[S_k(S_n - S_k) \cdot \mathbf{1}_{E_k}] + \mathbb{E}[(S_n - S_k)^2 \cdot \mathbf{1}_{E_k}]$$

but $S_k \cdot \mathbf{1}_{E_k} = (X_1 + \dots X_k) \cdot \mathbf{1}_{E_k}$ and $S_n - S_k = X_n + X_{n-1} + \cdots + X_{k+1}$ and, since X_k are independent, $S_k \cdot \mathbf{1}_{E_k}$ and $S_n - S_k$ are also independent, then

$$\mathbb{E}[S_k(S_n - S_k) \cdot \mathbf{1}_{E_k}] = \mathbb{E}[S_k \cdot \mathbf{1}_{E_k}] \cdot \cancelto{0}{\mathbb{E}[S_n - S_k]} = 0$$

so we can conclude that

$$\mathbb{E}[S_n^2 \cdot \mathbf{1}_{E_k}] = \mathbb{E}[S_k^2 \cdot \mathbf{1}_{E_k}] + \mathbb{E}[(S_n - S_k)^2 \cdot \mathbf{1}_{E_k}] \ge \mathbb{E}[S_k^2 \cdot \mathbf{1}_{E_k}] \tag{3}$$

now, substituting 3 in 2, we conclude that

$$\text{var}(S_n) \ge \sum_{k=1}^{n} \mathbb{E}[S_n^2 \cdot \mathbf{1}_{E_k}] \ge \sum_{k=1}^{n} \mathbb{E}[S_k^2 \cdot \mathbf{1}_{E_k}]$$

and, by definition, $|S_k(\omega)| \ge \varepsilon$ whenever $\mathbf{1}_{E_k}(\omega)$, therefore $|S_k^2(\omega)| \ge \varepsilon^2$ and, by 1, we have that

$$\text{var}(S_n) \ge \sum_{k=1}^{n} \mathbb{E}[\varepsilon^2 \cdot \mathbf{1}_{E_k}] = \varepsilon^2 \sum_{k=1}^{n} \mathbb{P}(E_k) = \varepsilon^2 \mathbb{P}(E)$$

$\dashv$

Theorem 2.4. (Kolmogorov's Two Series). Let $(X_n)_{n\geq 0}$ be independent random variables. If $\sum_n \mathbb{E}[X_n]$ and $\sum_n \operatorname{var}(X_n)$ both converges, then $\sum_n X_n$ converges almost surely.

Proof. Note first that, if we define $Y_k = X_k - \mathbb{E}[X_k]$, then Y_k are also independent, $\mathbb{E}[Y_n] = 0$ and $\operatorname{var}(Y_n) = \operatorname{var}(X_n)$. Now, since

$$\sum_n X_n = \sum_n (Y_n + \mathbb{E}[X_n]) = \sum_n Y_n + \sum_n \mathbb{E}[X_n]$$

and, by hypothesis, $\sum_n \mathbb{E}[X_n]$ converges, then $\sum_n X_n$ converges iff $\sum_n Y_n$ converges, then we can assume WLOG that $\mathbb{E}[X_n] = 0$.

We will show that, if $S_k = \sum_{i=1}^k X_k$, then $\mathbb{P}(S_k \text{ converges}) = 1$ or, equivalently, that S_k is almost surely Cauchy. In order to do that, note that

$$\max_{m,n\geq N} |S_n - S_m| \leq \max_{n\geq N} |S_n - S_N| + \max_{m\geq N} |S_n - S_M| \leq 2 \max_{n\geq N} |S_n - S_N|$$

then, by Kolmogorov's Inequality

$$\mathbb{P}\left(\max_{m,n\geq N} |S_n - S_m| > \varepsilon\right) \leq \mathbb{P}\left(\max_{n\geq N} |S_n - S_N| > \frac{\varepsilon}{2}\right) \leq \frac{4}{\varepsilon^2} \sum_{n=N+1}^{n} \operatorname{var}(S_n) \tag{4}$$

Now, note that, S_n is not Cauchy if, there is a $\varepsilon > 0$ rational such that, for all $N \in \omega$, there are $n, m \geq N$ with $|S_n - S_m| > \varepsilon$. Then, for S_n to be almost surely Cauchy it's sufficient that

$$\mathbb{P}\left(\bigcup_{\varepsilon\in\mathbb{Q}} \bigcap_{N\in\omega} \bigcup_{n,m\geq N} \{|S_n - S_m| > \varepsilon\}\right) \leq \sum_{\varepsilon\in\mathbb{Q}} \mathbb{P}\left(\bigcap_{N\in\omega} \bigcup_{n,m\geq N} \{|S_n - S_m| > \varepsilon\}\right) = 0$$

or that $\mathbb{P}\left(\bigcap_{N\in\omega} \bigcup_{n,m\geq N} \{|S_n - S_m| > \varepsilon\}\right) = 0$. But, if $E_N = \bigcup_{n,m\geq N} \{|S_n - S_m| > \varepsilon\}$, note that $E_n \supseteq E_{n+1}$, $\forall n \in \omega$, then, we have

$$\mathbb{P}\left(\bigcap_{N\in\omega} E_N\right) = \lim_{N\to\infty} \mathbb{P}(E_N)$$

and, since $E_N = \max_{m,n\geq N} |S_n - S_m| > \varepsilon$, by 4 we have that

$$\lim_{N\to\infty} \mathbb{P}(E_N) \leq \lim_{N\to\infty} \frac{4}{\varepsilon^2} \sum_{n=N+1}^{n} \operatorname{var}(S_n) = 0$$

where the summation is 0 because $\sum_n \operatorname{var}(S_n)$ converges. Then S_n is almost surely Cauchy. ⊣

Using Kolmogorov's Theorem we can prove the following Lemma due to Rademacher

Como usaremos somente $c_n = \frac{1}{n}$ na prova do teorema 2.5., e sabemos que a harmônica se relaciona com a densidade assintótica por meio do ln, talvez seja possível pular esse teorema

Theorem 2.5. (Rademacher). Let $(c_n)_{n\geq 0}$ be any sequence of reals. And let

$$A := \left\{ s \in 2^\omega : \sum_n (-1)^{s(n)} c_n \text{ converges} \right\}$$

Then $\lambda(A) = 1$ if $\sum_n c_n^2$ converges.

Proof. We will consider $X_n = (-1)^{s(n)} c_n$, where s_n is 0 and 1 with probability $\frac{1}{2}$. Now

$$\mathbb{E}[X_n] = \mathbb{E}[(-1)^{s(n)} c_n] = c_n \left(1 \cdot \frac{1}{2} + (-1) \cdot \frac{1}{2} \right) = 0$$

Therefore $\sum_n \mathbb{E}[X_n]$ converges. Now $\mathrm{var}(Y_n) = \mathbb{E}[Y_n^2] - \mathbb{E}[Y_n]^2$ and, since we have that

$$\mathbb{E}[Y_n^2] = c_n^2 \left(1 \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} \right) = c_n^2$$

then the second series became $\sum_n c_n^2$, which converges by assumption. And, by Kolmogorv's Two Series Theorem, $\lambda(A) = 1$. $\dashv$

With the previous theorem we can now give an inequality between $\mathrm{cov}(\mathcal{N})$ and $\mathfrak{rr}$.

Theorem 2.6. $\mathrm{cov}(\mathcal{N}) \leq \mathfrak{rr}$

Aqui estamos usando que $\mathrm{cov}(\mathcal{N})$ é o mesmo se $\mathcal{N}$ for escolhido em $\mathbb{R}$ ou no Espaço de Cantor 2^ω.

Proof. Since $\sum_n \frac{1}{n^2}$ converges absolutely, for any $p \in \mathrm{Sym}(\omega)$, since $\sum_n \frac{1}{p(n)^2}$ is just a rearrangement, then it also converges. Now, by Rademacher's Theorem, for $c_n = \frac{1}{p(n)}$, the set

$$A_p := \left\{ s \in 2^\omega : \sum_n (-1)^{s(n)}/p(n) \text{ diverges} \right\}$$

has measure zero. And, since $s \mapsto s \circ p$ is measure preserving in the Cantor Space 2^ω (Por que?), then $B_p = p[A_p]$ also has measure zero.

Now, let $C \subseteq \mathrm{Sym}(\omega)$ be s.t. $|C| < \mathrm{cov}(\mathcal{N})$. By definition of $\mathrm{cov}(\mathcal{N})$, we have that $\{B_p\}_{p \in C}$ cannot cover 2^ω, then there is some $s \in 2^\omega$ such that $\sum_n (-1)^{s(p(n))}/p(n)$ converges $\forall p \in C$. That is, the rearrangement of $\sum_n (-1)^{s(n)}/n$ will not diverge to $\pm\infty$ or oscillate.

Therefore, this proves that $\mathrm{cov}(\mathcal{N}) \leq \mathfrak{rr}_{io} = \mathfrak{rr}$. $\dashv$

2.5 Topology and Some Inequalities

In this subsection we will continue to present some inequalities between the rearrangement numbers and other well known cardinal characteristic, so we can try, for example, to see where the rearrangement numbers are in the Cichóns Diagram.

For the first inequality, we will need the following lemmas

Lemma 2.4. If $Y \subseteq X$ and X is meager, then Y is also meager.

Proof. Since X is meager, $X = \bigcup_{n\in\omega} X_n$, where each X_i is nowhere dense, i.e., $\text{int}\big(\overline{X_i}\big) = \emptyset$, $\forall i \in \omega$. Now, since $Y \subseteq X$, we have that

$$Y = Y \cap X = \bigcup_{n\in\omega} Y \cap X_n \tag{$*$}$$

but $Y \cap X_i \subseteq X_i$, which implies $\overline{Y \cap X_i} \subseteq \overline{X_i}$, then $\text{int}\big(\overline{Y \cap X_i}\big) \subseteq \text{int}\big(\overline{X_i}\big) = \emptyset$, i.e., $Y \cap X_i$ are nowhere dense, therefore by $(*)$ Y is also meager. $\dashv$

Lemma 2.5. The subspace $\text{Sym}(\omega)$ of ω^ω of all permutations of ω is homeomorphic to ω^ω

Proof. Uma possível demonstração usa o Teorema de Alexandrov-Uryhson, cujos pré-requisitos estão far beyond what I know. Outra possível demonstração é a construção direta do homeomorfismo como eles fazem no artigo do número de rearranjo, mas não é muito intuitiva. $\dashv$

Corollary 2.1. Let $\mathcal{M}_X$ be the set of meager sets in the space X. If $X \cong Y$, then

$$\text{non}(\mathcal{M}_X) = \text{non}(\mathcal{M}_Y)$$

Proof. By symmetry, it's sufficient to show that, if $f : X \cong Y$, then A is meager in X implies $f(A)$ is meager in Y. It's a classical result in topology that, if f is a homeomorphism, then it commutes with cl and int. Therefore, if $A = \bigcup_{n\in\omega} A_n$, with $\text{int}\big(\overline{A_n}\big) = \emptyset$, then

$$f(A) = f\left(\bigcup_{n\in\omega} A_n\right) = \bigcup_{n\in\omega} f(A_n)$$

with $\text{int}\left(\overline{f(A_n)}\right) = f\big(\text{int}\big(\overline{A}\big)\big) = \emptyset$, i.e., $f(A)$ is also meager. $\dashv$

Theorem 2.7.

$$\mathfrak{rr} \leq \textbf{non}(\mathcal{M})$$

Proof. Its sufficient to prove that any non-meager set $C \subseteq \text{Sym}(\omega)$ has, given some series, a permutation that corrupts it. Given a c.c. series $\sum_n a_n$ of reals, consider

$$X(a_n) := \left\{ p \in \text{Sym}(\omega) : \sum_n a_{p(n)} \neq \sum_n a_n \right\}$$

we need to show that $C \cap X(a_n) \neq \emptyset$, then, its sufficient to show that $X(a_n)$ is comeager, since, if it was empty, then $C \subseteq X(a_n)^c$, but, by Lemma 2.4, C would also be meager.

To show $X(a_n)$ is comeager, its sufficient to prove the stronger claim that

$$Y(a_n) := \left\{ p \in \mathrm{Sym}(\omega) : \sum_n a_{p(n)} = \infty \right\}$$

is the intersection of countable many dense open sets, i.e. $Y = \bigcap_{n\in\omega} D_n$ with D_n open and denses. In fact, first of all, since D_n is open, D_n^c is closed, then, if $\mathrm{int}(\overline{D_n^c}) = \mathrm{int}(D_n^c) \neq \emptyset$, there would be an open $U \subseteq D_n^c$, so $U \cap D_n = \emptyset$, contradicting that D_n is dense, therefore D_n is nowhere dense for all $n \in \omega$. Now, since $Y^c = \bigcup_{n\in\omega} D_n^c$, Y^c is meager and, since $Y(a_n) \subseteq X(a_n)$, then $X(a_n)^c \subseteq Y(a_n)^c$, so, by Lemma 2.4, $X(a_n)^c$ would be meager too.

So, since we have that

$$Y(a_n) = \bigcap_{k\in\omega} \bigcup_{m\geq k} \left\{ p \in \mathrm{Sym}(\omega) : \sum_{n=0}^{m} a_{p(n)} \geq k \right\}$$

its sufficient to prove that

$$D_k := \bigcup_{m\geq k} \left\{ p \in \mathrm{Sym}(\omega) : \sum_{n=0}^{m} a_{p(n)} \geq k \right\}$$

is dense and open. If $p \in D_k$, $\sum_{n=0}^{m} a_{p(n)} \geq k$ for some $m \geq k$. Then, given any $q \in \mathcal{N}(p \restriction_{m+1})$ extending p until m, we have that

$$\sum_{n=0}^{m} a_{q(n)} = \sum_{n=0}^{m} a_{p(n)} \geq k$$

So $\mathcal{N}(p \restriction_{m+1}) \subseteq D_k$, then its open. Now, we will prove its dense, i.e., for all $p \in \omega^{<\omega}$, we have $D_k \cap \mathcal{N}(p) \neq \emptyset$. Given $p \in \omega^{<\omega}$, extend it with positive terms to p' with domain $\geq k$ such that $\sum_{n=0}^{m} a_{p'(n)} \geq k$, which exists, since $\sum_{n=0}^{m} a_n^+ = \infty$. $\dashv$

Its hard to see how to change the argument so that $\sum_n a_{p(n)}$ could converge or diverge to $\pm\infty$. In fact, you can't, this is not a defect of the argument, these large rearrangement numbers can consistently be larger than $\mathbf{non}(\mathcal{M})$.

For the sake of the argument we will now prove the following inequality, which will be superseded by a stronger one in the next subsection.

Theorem 2.8.

$$\mathfrak{rr}_{fi} \geq \mathbf{cov}(\mathcal{M})$$

Proof. Given $p \in \mathrm{Sym}(\omega)$ consider the set

$$\begin{aligned} A_p &:= \{g \in \mathrm{Sym}(\omega) : g[M] = p[M] \text{ for infinitely many } M\} \\ &= \bigcap_{m\in\omega} \bigcup_{n\geq m} \{g \in \mathrm{Sym}(\omega) : g[n] = p[n]\} \end{aligned}$$

as we saw in 2.7, to show A_p is comeager, its sufficient to show that

$$U_m := \bigcup_{n\geq m} \{g \in \mathrm{Sym}(\omega) : g[n] = p[n]\} = \bigcup_{n\geq m} B_n$$

is open and dense. In fact, given $g \in U_m$, then $g \in B_k$ for some $k \geq m$, so $\mathcal{N}(g(0), \ldots, g(k)) \subseteq U_m$, then it's open. Now, given $p \in \mathrm{Sym}(\omega)$, p is injective, so, by The Mix Lemma 2.1, it can be extended to $p' \in \mathrm{Sym}(\omega)$ s.t. $p[n] = p'[n]$ for infinitely many $n \in \omega$, so $p' \in \mathcal{N}(p(0), \ldots, p(k)) \cap U_m$, for any $k \in \omega$, so it's dense.

Now, suppose toward a contradiction that $\mathfrak{rr}_{fi} < \mathbf{cov}(\mathcal{M})$ and let C witness this. Assume WLOG that $\mathrm{id} \in C$. Since $\bigcap_{p\in C} A_p$ is the intersection of fewer than $\mathbf{cov}(\mathcal{M})$ comeager sets, it's nonempty, so let g be a member of it.

Then, for any c.c. series $\sum_n a_n$, let $p \in C$ be such that $\sum_n a_{p(n)}$ converges to a different number or diverges to infinity. Since $g \in A_p$, $g[M] = p[M]$ for infinitely many M, and also $g[M] = M$ for infinitely many, since $\mathrm{id} \in C$. As a result, the same argument as in 2.3 shows that $\sum_n a_{g(n)}$ diverges by oscillation, so $\{g\}$ witnesses that $\mathfrak{rr}_o = 1$, a contradiction, since it's uncountable. $\dashv$

2.6 Advanced padding

The key idea in the padding method was to spread out the nonzero terms in a series so far that the permutations under consideration do not change their relative order up to finitely many exceptions. Now we will introduce a cardinal characteristic intended to capture this idea.

Definition 2.2. Let $A \subseteq \omega$

- A is preserved by $p \in \mathrm{Sym}(\omega)$ if p doesn't change the relative order of members of A, except for finitely many elements. If A is not preserved by p we say it's jumbled by p.
- A conserving family is a family of permutations such that there is an infinite set A that is preserved by every p in the family.
- The jumbling number $\mathfrak{j}$ is the greatest cardinality of a conserving family.

Note that, if $S = \{q_n : n \in \omega\} \subseteq \mathrm{Sym}(\omega)$, let $C = \{p_m = q_m^{-1} : m \in \omega\}$, then, by doing the construction of the fast growing function $\ell : \omega \to \omega$ as in 2.2, we guarantee that $p_m(\ell(m)) < p_m(\ell(m+1)) < \ldots$, which means $A = \{\ell(n) : n \in \omega\}$ is not jumbled by any $p \in C$, then, since $\aleph_0 < \mathfrak{j} \leq |\mathrm{Sym}(\omega)| = \mathfrak{c}$, it's a cardinal characteristic of the continuum. This proof is not necessary, since Theorem 2.10 will guarantee that (but the argument is way harder. way more powerful, tho).

Roughly, $\mathfrak{j}$ is defined to be the largest cardinal for which the argument in Theorem 2.2 remains valid. Just consider that for any fast growing such function ℓ we can associate the set $A = \{\ell(1), \ell(2), \ldots\}$ and vice-versa. We will see how to use this in the proof of the next Theorem.

Theorem 2.9.

$$\mathfrak{j} \leq \mathfrak{rr}$$

Proof. The idea is to use the same translation as in the proof that $\mathfrak{j}$ is uncountable. Let $C \subseteq \mathrm{Sym}(\omega)$, with $|C| < \mathfrak{j}$. We shall find a c.c. series that is not corrupted by C. Let $\sum_n b_n$ be any c.c. series and, since $|C| < \mathfrak{j}$ let $A \subseteq \omega$ be the set that is not jumbled by *the inverses of all permutations* in C. Then, using A as the growing function, let

$$a_n := \begin{cases} b_k, & \text{if } n \text{ is the } k^{\text{th}} \text{ element of } A \\ 0, & \text{if } n \notin A \end{cases}$$

If $p \in C$, then A is preserved by p^{-1}, then the order of almost all nonzero terms of $\sum_n a_n$ and $\sum_n a_{p(n)}$ are the same, therefore C does not corrupt $\sum_n a_n$. ⊣

In order to bound $\mathfrak{rr}$ with a well known cardinal characteristic, we will prove the following theorem

Theorem 2.10. $\mathfrak{j} \leq \mathfrak{b}$, therefore, $\mathfrak{b} \leq \mathfrak{rr}$.

Sketch. To prove $\mathfrak{b} \leq \mathfrak{j}$, we will take $C \subseteq \mathrm{Sym}(\omega)$ s.t. $|C| < \mathfrak{b}$ and prove it cannot be a jumbling family. Since $|C| < \mathfrak{b}$, we need to associate for each $p \in C$ a function $f_p : \omega \to \omega$ so we could have a function $g : \omega \to \omega$ that dominates f_p for all permutations $p \in C$ and, using g, we will define a set $A \subseteq \omega$ that is not jumbled by C.

Fixing $p \in C$, we want to define f_p such that, for some $g \geq^* f_p$ and A we will have, p will preserve the order of most elements in A. Let's then define f_p as an *Stabilization Point*, i.e, $f_p(n)$ is s.t. from this point on, every number will have the image over p greater than $p(1), p(2), \dots, p(n)$.

As Figure 5 shows, given $1, 2, 3, \dots, n$, and labeling them $b_1, b_2, \dots, b_n$ such that $p(b_1) < p(b_2) < \cdots < p(b_n)$ (points represented in purple), we know that for each number between the images of $1, 2, \dots, n$ there will be some number y_i that will go there (numbers in blue), then, if we run out all of them, any other number after those ones will necessarily be greater than the images of $1, 2, \dots, n$ over p, then we just need to choose $f_p(n)$ (in red) far enough so he can surpass all these finite elements.

Then, if we define $a_0 \in \omega$ and $a_{n+1} \in \omega$ such that $a_{n+1} > g(a_n)$, it's easy to see that $A = \{a_i\}_{i \in \omega}$ is preserved by C.

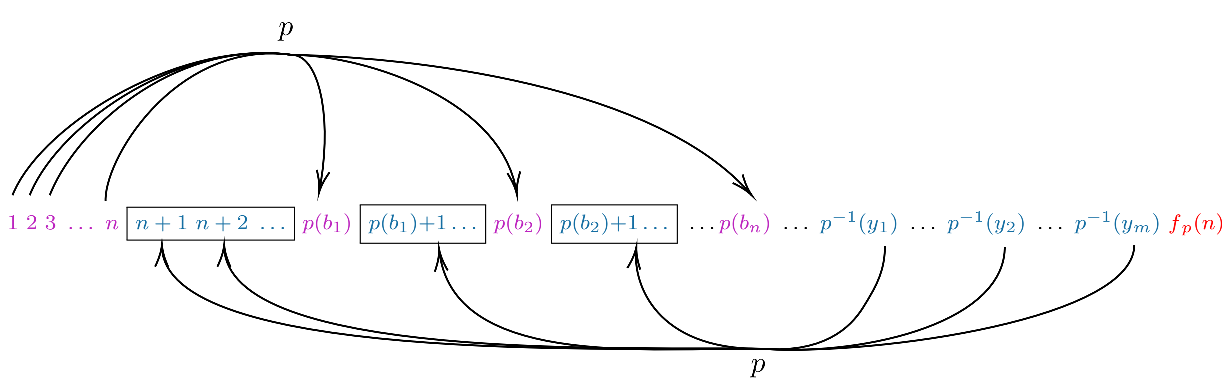

Fig. 5: Visualization of $f_p(n)$

Proof. Let $C \subseteq \mathrm{Sym}(\omega)$ be s.t. $|C| < \mathfrak{b}$ and, for each $p \in C$, define $f_p : \omega \to \omega$ a function such that $\forall n \in \omega$, $\forall x \leq n$ and $\forall y \geq f_p(n)$, $p(x) < p(y)$, which we saw it's not hard to find. Then, since $\{f_p\}_{p\in C}$ has less than $\mathfrak{b}$ functions, there is a $g : \omega \to \omega$ such that $g \geq^* f_p$, $\forall p \in C$. Let a_0 be any natural and a_{n+1} a natural such that $a_{n+1} > g(a_n)$. Then, for fixed $p \in C$, let $k \in \omega$ be such that $g(n) \geq f_p(n)$, $\forall n \geq k$, then, for all $k \leq a_r < a_s$

$$f_p(a_r) \leq g(a_r) < a_{r+1} \leq a_s$$

and, by definition of f_p, $p(a_r) < p(a_s)$, therefore $A = \{a_n : n \in \omega\}$ is eventually preserved by any $p \in C$, therefore $\mathfrak{j} \leq \mathfrak{b}$. $\dashv$

Indeed, we have that $\mathfrak{j} = \mathfrak{b}$, which is proved in rearrangement number article using another characterization of $\mathfrak{b}$ showed in Blass chapter in the Handbook. But, since $\mathfrak{j}$ isn't mentioned after this proof, I think it's more like a fun fact, therefore I will not add the proof here.

Theorem 2.11.

$$\mathfrak{d} \leq \mathfrak{rr}_{fi}$$

Proof. Let $C \subseteq \mathrm{Sym}(\omega)$ be such that $|C| < \mathfrak{d}$, we want to find a $\sum_n a_n$ such that none of the permutations in C make it diverge to $\pm\infty$ or converge to a different number.

Consider the same association $p \mapsto f_p$ as in the previous proof, but use $p \mapsto f_{p^{-1}}$, instead. Then $f_p(n) > n$ for every $n \in \omega$, and $\forall x \leq n$ and $y \geq f_p(n)$ we have $p^{-1}(x) < p^{-1}(y)$.

Since $|C| < \mathfrak{d}$ there is some $g \in \omega^\omega$ that is not eventually dominated by any f_p, with $p \in C$. By increasing its values if necessary we can assume that g is strictly increasing and $g(n) > n$, then

$$0 < g(0) < g(g(0)) < \cdots < g^k(0) < g^{k+1}(0) < \ldots$$

which we will use to use the technique of padding with zeros. Fix any c.c. series $\sum_n b_n$ and consider

$$a_n = \begin{cases} b_k, \text{ if } n = g^k(0); \\ 0, \text{ otherwise.} \end{cases}$$

We claim that $\sum_n a_n$ is such series we were looking for.

Given $p \in C$, since g is not dominated by f_p, there are infinitely many $n_1, n_2, \dots \in \omega$ such that $f_p(n_i) < g(n_i)$, $\forall i \geq 1$. Fixing some such n, let k be the smallest integer such that $n < g^k(0)$, then $g^{k-1}(0) \leq n$ and $g(n) < g^{k+1}(n)$. Therefore, since $f_p(n) > n$, we have

$$g^{k-1}(0) \leq n < f_p(n) < g(n) < g^{k+1}(0)$$

by our definition of f_p, since $0, g(0), \dots, g^{k-1}(0) \leq n$ and $g^{k+1}(0), \dots \geq f_p(n)$, then

$$p^{-1}(0), p^{-1}(g(0)), \dots, p^{-1}(g^{k-1}(0)) < p^{-1}(g^{k+1}(0)), \dots$$

In particular, since $p^{-1}(g^r(0))$ is the position where b_r appears in $\sum_n a_{p(n)}$, then all of $b_0, b_1, \dots, b_{k-1}$ occur before all of $b_{k+1}, b_{k+2}, \dots$ and, since b_k can occur anywhere, we know that the partial sum of $\sum_n a_{p(n)}$ differs from $\sum_{n=0}^k b_n$ by at most $|b_k|$ (more precisely, if we let

$$N(k) = \max\{p^{-1}(0), p^{-1}(g(0)), \dots, p^{-1}(g^{k-1}(0))\}$$

then $\sum_{n=0}^{N(k)} a_{p(n)}$ differs from $\sum_{n=0}^{k} b_n$ by at most $|b_k|$).

When we let $n = n_i$, with $i \to \infty$, we have $k = k_i \to \infty$ too. Now, since $|b_{k_i}| \to 0$, we have that $\sum_{n=0}^{N(k_i)} a_{p(n)} \to \sum_n b_n = \sum_n a_n$, then the partial sums of $\sum_n a_{p(n)}$ has a subsequence that converges to $\sum_n a_n$, which means it can't diverge to $\pm\infty$ or converge to a different finite sum. $\dashv$

3 The Density Cardinal

3.1 Introduction

Motivação

For $A \subseteq \omega$, let $d_n(A) = \frac{|A \cap n|}{n}$ and

$$\underline{d}(A) = \liminf_{n\to\infty} d_n(A)$$
$$\overline{d}(A) = \limsup_{n\to\infty} d_n(A)$$

If $\underline{d}(A) = \overline{d}(A)$ we call the common value $d(A)$ the (*asymptotic*) *density* of A. Note that, when A has asymptotic density, we have that $d(A) \in [0, 1]$. On the other hand, analogously to Riemann's Rearrangement Theorem, we can prove the following theorem:

Theorem 3.1. Given an infinite and coinfinite set A, we have

- For every $r \in [0, 1]$, there is a $p \in \mathrm{Sym}(\omega)$ such that $d(p[A]) = r$;
- There is a $p \in \mathrm{Sym}(\omega)$ such that $p[A]$ does not have asymptotic density

Proof. Given $r \in (0, 1)$, let $p_1 = 1$ and $B_1 = \{1\}$. Now, let q_1 be the least $N > p_1$ such that $d_N(B_1) < r$. Now, define p_i, B_i and q_i, for $i > 1$ recursively as follows:

Let p_n be the least $N > q_{n-1}$ such that $B_n = B_{n-1} \cup \{q_{n-1}+1, q_{n-1}+2, \dots, N\}$ has $d_N(B_n) > r$. And q_n be the least $N > p_n$ such that $d_N(B_n) < r$.

Now, let $B = \bigcup_{i\geq 1} B_i = \{b_1, b_2, b_3, \dots\}$ with $b_1 < b_2 < \dots$ and $B^c = \{\tilde{b_1}, \tilde{b_2}, \dots\}$ also in order. Analogously, let $A = \{a_1, a_2, \dots\}$ and $A^c = \{\tilde{a_1}, \tilde{a_2}, \dots\}$ both in order. We can then define $p \in \mathrm{Sym}(\omega)$ such that $p(a_i) = b_i$ and $p(\tilde{a_i}) = \tilde{b_i}$, which is obviously a permutation in ω.

With this we have that, since p_i is the least N with $d_N(p[A]) > r$, for any $i > 1$

$$d_{p_i-1}(p[A]) < r < d_{p_i}(p[A])$$

therefore

$$\begin{aligned}
0 < d_{p_i}(p[A]) - r &< d_{p_i}(p[A]) - d_{p_i-1}(p[A]) \\
&= \frac{|p[A] \cap p_i|}{p_i} - \frac{|p[A] \cap (p_i - 1)|}{p_i - 1} \\
&< \frac{|p[A] \cap p_i| - |p[A] \cap (p_i - 1)|}{p_i - 1}
\end{aligned}$$

and, since the numerator of the RHS is either 0 or 1, and $p_i - 1 \to \infty$, then, when $i \to \infty$, the RHS converges to 0, therefore $d_{p_i}(p[A]) \to r$. Similarly, since $d_{q_i}(p[A]) < r < d_{q_i-1}(p[A])$, we have that $d_{q_i}(p[A]) \to r$. And, because

$$d_{q_i}(p[A]) \leq d_{q_i-1}(p[A]) \leq \cdots \leq d_{p_i-2}(p[A]) \leq d_{p_i-1}(p[A]) \leq d_{p_i}(p[A])$$

then $d_n(p[A]) \to r$.

The cases for $p \in \mathrm{Sym}(\omega)$ s.t. $d(p[A])$ is 0 or 1, and where $p[A]$ does not have asymptotic density are analogous. ⊣

Lemma 3.1. Let (a_n) be a sequence in ω satisfying that there exists $n_0 \in \omega$ such that for all $n \geq n_0$, we have that $a_{n+1} - a_n > 2^n$. Then $A = \{a_n\}_{n \geq 0}$ is infinite-coinfinite and $d(A) = 0$.

Proof. It's clear that A is infinite-coinfinite. Now, since (a_n) is increasing for $n \geq n_0$, we have that

$$a_{n+1} > a_n + 2^n > a_{n_0} + 2^n$$

therefore, if $k \in \omega$ is s.t. $a_{n_0} + 2^{n-1} \leq k \leq a_{n_0} + 2^n < a_{n+1}$, then $A \cap k \subseteq \{a_i : i \leq n\}$, for $n \geq n_0$, and we will have that

$$d_k(A) = \frac{|A \cap k|}{k} \leq \frac{n+1}{k} \leq \frac{n+1}{a_{n_0} + 2^{n-1}}$$

which converges to 0 when $k \to \infty$, therefore $d(A) = 0$. ⊣

Theorem 3.2. $\mathfrak{dd}$ is uncountable.

Proof. Não sei porque eu não completei isso ainda, é um corolário direto do lema anterior ⊣

3.2 Characterization of $\mathfrak{dd}$

The main theorem of this subsection is the characterization $\mathfrak{dd} = \mathbf{non}(\mathcal{M})$, where the latter is a well known cardinal characteristic. To prove this using GT, we will find relational systems $\mathcal{A}_1$, $\mathcal{A}_2$ and $\mathcal{B}$ s.t. $\mathcal{A}_1 \leq_T \mathcal{B} \leq_T \mathcal{A}_2$ and $\mathfrak{d}(\mathcal{A}_1) = \mathfrak{d}(\mathcal{A}_2) = \mathbf{non}(\mathcal{M})$ and $\mathfrak{d}(\mathcal{B}) = \mathfrak{dd}$.

Its well known that $\mathcal{A}_2 = (\mathcal{M}, 2^\omega, \not\ni)$ works, where $\mathcal{M}$ are the meager sets of 2^ω. And, for $\mathfrak{dd}$, we can take $\mathcal{B} = (D, \mathrm{Sym}(\omega), R)$, where D are the infinite-coinfinite substets of ω with density, and xRp if $d(x) \neq d(p(x))$, as expected.

We will first prove the easier inequality $\mathcal{B} \leq_T \mathcal{A}_2$, you should try to find a reduction for $\mathcal{A}_2 \leq_T \mathcal{B}$, you will see that's not too easy (in fact it's impossible), this is why we will later find other relational system $\mathcal{A}_1$ to work with the same dominating number.

Lemma 3.2.

$$(D, \mathrm{Sym}(\omega), R) \leq_T (\mathcal{M}, 2^\omega, \not\ni)$$

Aqui usamos que ω^ω é homeomorfo a $\mathrm{Sym}(\omega)$ em 2^ω

Proof. We need to define $\varphi_- : D \to \mathcal{M}$ and $\varphi_+ : \mathrm{Sym}(\omega) \to \mathrm{Sym}(\omega)$ such that, for all $A \in D$ and $p \in \mathrm{Sym}(\omega)$, if $p \notin \varphi_-(A)$, then $d(A) \neq d(\varphi_+(p)(A))$. Lets take $\varphi_+ = \mathrm{id}$ as simple as possible, so we need $\varphi_-(A)$ such that, if $p \notin \varphi_-(A)$, then $d(A) \neq d(p(A))$.

We can't choose p such that $d(p(A)) = 1$ (as in $\sum_n a_{p(n)} = \infty$), since $\mathfrak{dd}$ is the analogous cardinal of $\mathfrak{rr}'$ and not $\mathfrak{rr}$. Then we will choose another approach:

Lets try to force $d(p(A))$ diverge by oscillation, so $d_{n_k}(p(A)) \to 0$ and $d_{m_k}(p(A)) \to 1$ for two different subsequences. Then, $\forall \varepsilon > 0$ and $\ell \in \omega$, there are $n_0, n_1 \geq \ell$ such that $d_{n_0}(p(A)) < \varepsilon$ and $d_{n_0}(p(A)) > 1 - \varepsilon$. Then, a natural choice for $\varphi_-(A)^c$ would be

$$\begin{aligned}
\varphi_-(A)^c &:= \{p \in \mathrm{Sym}(\omega) : \forall \varepsilon > 0, \forall \ell \in \omega, \exists n_0, n_1 \geq \ell, d_{n_0}(p(A)) < \varepsilon \text{ and } d_{n_1}(p(A)) > 1 - \varepsilon\} \\
&= \left\{p \in \mathrm{Sym}(\omega) : \forall k, \ell \in \omega, \exists n_0, n_1 \geq \ell, d_{n_0}(p(A)) < \frac{1}{k} \text{ and } d_{n_1}(p(A)) > 1 - \frac{1}{k}\right\} \\
&= \bigcap_{k,\ell \in \omega} \left\{p \in \mathrm{Sym}(\omega) : \exists n_0, n_1 \geq \ell, d_{n_0}(p(A)) < \frac{1}{k} \text{ and } d_{n_1}(p(A)) > 1 - \frac{1}{k}\right\} \\
&= \bigcap_{k,\ell \in \omega} A_{k,\ell}
\end{aligned}$$

as we saw in the proof of theorem 2.7, to show that $\varphi_-(A) \in \mathcal{M}$, it's sufficient to prove that $\varphi_-(A)^c$ is the intersection of countably many dense sets, so we will show that each $A_{k,\ell}$ is open and dense. If $p \in A_{k,\ell}$, lets $m = \max\{n_0, n_1\}$, then, for $q \in \mathcal{N}(p\restriction_{m+1})$, clearly $d_{n_0}(q(A)) = d_{n_0}(p(A)) < \frac{1}{k}$ and $d_{n_1}(q(A)) = d_{n_1}(p(A)) > 1 - \frac{1}{k}$, so $q \in A_{k,\ell}$, and then $\mathcal{N}(p\restriction_{m+1}) \subseteq A_{k,\ell}$, so it's open.

To show it's dense, given $p \in \mathrm{Sym}(\omega)$ and $m \in \omega$, we need to prove $\mathcal{N}(p\restriction_{m+1}) \cap A_{k,\ell} \neq \emptyset$. Take $q \in \mathcal{N}(p\restriction_{m+1})$ and extend it randomly until ℓ. Now, extend q for $r \geq \ell$ such that $q(r) \notin \omega \setminus A$ until $d_n(q(A)) < \frac{1}{k}$, after this choose $q(r) \in A$ such that $d_n(q(A)) > 1 - \frac{1}{k}$. Then $q \in A_{k,\ell}$. ⊣

Definition 3.1. Let $h \in \omega^\omega$ with $|h(n)| \geq 1$, $\forall n \in \omega$. A function $\phi : \omega \to [\omega]^{<\omega}$ is an h-slalom if $|\phi(n)| \leq h(n)$, $\forall n \in \omega$. We denote by Φ_h the collection of all h-slaloms.

Now, to prove the other tukey inequality, let's take $\mathcal{A}_1 = (\omega^\omega, \Phi_h, \in^\infty)$, where $f \in^\infty \phi$ if $f(n) \in \phi(n)$ for infinitely many n, and use a result due to Bartoszyński (Combinatorial aspects of measure and category):

Theorem 3.3. (Bartoszyński) For any $h \in \omega^\omega$ satisfying $|h(n)| \geq 1$, $\forall n \in \omega$

$$\mathbf{non}(\mathcal{M}) = \mathfrak{d}(\omega^\omega, \Phi_h, \in^\infty);$$
$$\mathbf{cov}(\mathcal{M}) = \mathfrak{b}(\omega^\omega, \Phi_h, \in^\infty).$$

It's possible to show that $(\omega^\omega, \Phi_h, \in^\infty)$ and $(\mathcal{M}, 2^\omega, \not\ni)$ are not Tukey equivalent, so the approach to prove $\mathfrak{dd} \geq \mathbf{non}(\mathcal{M})$ changes a lot from the proof of Lemma 3.2 and it's way trickier.

Lemma 3.3. For any $h \in \omega^\omega$ growing fast enough (as equation (5) in the sketch below)

$$(\omega^\omega, \Phi_h, \in^\infty) \leq_T (D, \mathrm{Sym}(\omega), R)$$

Sketch. Before giving the formal proof, let's sketch what is the intuitive idea behind. We have some $h \in \omega^\omega$ and we want $\varphi_- : \omega^\omega \to D$ and $\varphi_+ : \mathrm{Sym}(\omega) \to \Phi_h$ such that

$$\varphi_-(g)Rp \Rightarrow g \in^\infty \varphi_+(p),\ \forall g \in \omega^\omega,\ \forall p \in \mathrm{Sym}(\omega)$$

Since $\varphi_-(g)Rp$ means that $d(\varphi_-(g)) \neq d(p(\varphi_-(g)))$, we will go for the contrapositive, i.e., assume there is $n_0 \in \omega$ such that $g(n) \notin \varphi_+(p)(n)$, $\forall n \geq n_0$ and force that $d(\varphi_-(g)) = d(p(\varphi_-(g)))$.

Now, since we have the freedom to choose h, let's choose it so we could prove $d(\varphi_-(g)) = 0$. If we could take $\varphi_+(p)$ such that $2^n \subseteq \varphi_+(p)(n)$, then, since $g(n) \notin \varphi_+(p)(n)$, $\forall n \geq n_0$, we have $g(n) \geq 2^n$, which implies that, if $\varphi_-(g)$ grows as fast as g, then $d(\varphi_-(g)) = 0$. Indeed, we could try to take $\varphi_-(g) = \{2^n\}_{n\in\omega}$, but then we lost the control we have, because we can no longer use our hypothesis $g(n) \notin \varphi_+(p)(n)$, so we will use a clever definition that depends on g:

Let $\varphi_-(g) = \{a_n\}_{n\in\omega}$ be such that (a_n) is any increasing sequence, for $n < n_0$; and $a_{n+1} = g(a_n)$, for $n \geq n_0$. Then, since $g(n) \geq 2^n$, $\forall n \geq n_0$, it implies that

$$a_{n+1} = g(a_n) \geq 2^{a_n} \geq 2^n, \text{ for } n \geq n_0$$

so we have that $d(\varphi_-(g)) = 0$ as desired.

Now, it remains to choose $h \in \omega^\omega$ such that $2^n \in \varphi_+(p)(n)$ and $d(p(\varphi_-(g))) = 0$. We want that p sparse the elements of $\varphi_-(g)$ as much as g, in particular, it would be good if $p(a_{n+2}) \geq a_{n+1} \geq 2^n$, for all $n \geq n_0$, in this case $d(p(\varphi_-(g))) = 0$. To do this, we need that

$$p(a_{n+1}) \neq 0, 1, \ldots, a_n, \text{ i.e., } a_{n+1} \neq p^{-1}(0), p^{-1}(1), \ldots, p^{-1}(a_n)$$

So let's choose

$$\varphi_+(p) = 2^n \cup \{p^{-1}(0), p^{-1}(1), \ldots, p^{-1}(n)\}$$

then, $a_{n+1} = g(a_n) \notin \varphi_+(p)(a_n)$, $\forall n \geq n_0$ implies $p(a_{n+1}) \geq a_n = g(a_{n-1}) \geq 2^{n-1}$. So the natural choice for h is any $h \in \omega^\omega$ such that

$$h(n) \geq 2^n + n + 1 = |\varphi_+(p)| \tag{5}$$

Proof. Let $h \in \omega^\omega$ be such that $h(n) \geq 2^n + n + 1$, $\forall n \in \omega$. Choose $\varphi_+(p)(n) = 2^n \cup \{p^{-1}(k) : k \leq n\}$, then clearly $\varphi_+(p) \in \Phi_h$. Now, assume there is some $n_0 \in \omega$ such that $g(n) \notin \varphi_+(p)(n)$, and therefore $g(n) \geq 2^n$, $\forall n \geq n_0$. Define $\varphi_-(g) = \{a_n : n \in \omega\}$ where (a_n) is any increasing sequence for $n < n_0$, and $a_{n+1} = g(a_n)$, for $n \geq n_0$. Since $a_{n+1} = g(a_n) \notin \varphi_+(p)(a_n)$, $\forall n \geq n_0$, then $a_{n+1} \neq p^{-1}(0) \ldots, p^{-1}(a_n)$, i.e., $p(a_{n+1}) \neq 0, 1, \ldots, a_n$, so $p(a_{n+1}) > a_n \geq 2^{n-1}$, which means $d(\varphi_-(g)) = 0 = d(p(\varphi_-(g)))$. $\dashv$

With such characterization in hands, and given that we have defined $\mathfrak{dd}$ as an analogous cardinal to $\mathfrak{rr}$, it's a natural question to ask if $\mathfrak{rr} = \mathbf{non}(\mathcal{M})$ also. In fact, it's far from clear if this is true.

The first reason is that, in some sense, $\mathfrak{dd}$ is not actually the analogous cardinal to $\mathfrak{rr}$, since c.c. series corresponds to sets with density strictly between 0 and 1, rather than p.c.c. series diverging to $\pm\infty$. The following table summarizes this correspondence:

p.c.c. serie $\sum_n a_n$	infinite-coinfinite set A
c.c. series	$d(A) \in (0,1)$
$\sum_n a_n = \pm\infty$	$d(A) \in \{0,1\}$
$\sum_n a_n$ diverges by oscillation	$d(A)$ is undefined

where $\sum_n a_n$ is p.c.c. (potentially conditionally convergent) if there is some $p \in \mathrm{Sym}(\omega)$ such that $\sum_n a_{p(n)}$ is c.c. In other words, it's c.c. without the condition that the original series converges to a finite number. With this in mind, we can define the actual cardinal analogous of $\mathfrak{dd}$:

Definition 3.2. $\mathfrak{rr}$ is the smallest cardinality of any family $C \subseteq \mathrm{Sym}(\omega)$ such that, for every p.c.c. series $\sum_n a_n$ of reals, there is some $p \in C$ for which $\sum_n a_{p(n)}$ no longer converges to the same limit.

using the analogous proofs for $\mathfrak{rr}$, it's not hard to see that $\mathfrak{rr} \leq \mathfrak{rr}' \leq \mathbf{non}(\mathcal{M})$. In fact, we have:

Theorem 3.4. $\mathfrak{rr}' = \mathbf{non}(\mathcal{M})$

whose proof is as the proof for $\mathfrak{dd}$, since it's sufficient to show that $(\omega^\omega, \Phi_h, \in^\infty) \leq_T (\text{p.c.c.}, \mathrm{Sym}(\omega), R)$

Proof. Let φ_+ be the same, i.e., $\varphi_+(p)(n) = 2^n \cup \{p^{-1}(k) : k \leq n\}$, then, if $g \in \omega^\omega$ is such that $g(n) \notin \varphi_+(p)(n)$ for all $n \geq n_0$, for some $n_0 \in \omega$, then, let $A = \{a_n : n \in \omega\}$ (which was $\varphi_-(g)$ in the previous proof) and $B = \{b_0 < b_1 < \dots\} = \omega \backslash A$. We define $\varphi_-(g) = c_k$, where

$$c_k = \begin{cases} \frac{1}{n}, \text{ if } k = b_n \\ -\frac{1}{n}, \text{ if } k = a_n \end{cases}$$

clearly $\sum_k c_k$ is p.c.c. and, since $a_n \geq 2^{n-1}$, we have $\sum_k c_k = \infty$. Also, since by our choice of h, we have $p(a_{n+2}) \geq 2^n$, then $\sum_k c_{p(k)} = \infty$ for all p (por que??) $\dashv$

3.3 Variations on $\mathfrak{dd}$

Motivated by the fact that $\mathfrak{rr}' = \mathbf{non}(\mathcal{M}) = \mathfrak{dd}$, we will generalize the definition of $\mathfrak{dd}$ as following:

Definition 3.3. Let $X, Y \subseteq \mathbf{all} = [0,1] \cup \{\mathbf{osc}\}$ where $X \neq \emptyset$ and for all $x \in X$ there is some $y \in Y$ such that $y \neq x$. The (X,Y)-*density number* $\mathfrak{dd}_{X,Y}$ is the smallest cardinality of a family $C \subseteq \mathrm{Sym}(\omega)$ such that for every infinite-coinfinite set $A \subseteq \omega$ with $d(A) \in X$, there is some $p \in C$ such that $d(p[A]) \in Y$ and $d(p[A]) \neq d(A)$.

The first two conditions are to guarantee that $\mathfrak{dd}_{X,Y}$ is well-defined, since we need $d(A) \in X$ (then $X \neq \emptyset$) and, if $d(A) = x \in X$, we need some $y \in Y$, such that $d(p[A]) = y \neq x$.

With this notation, it's easy to see that $\mathfrak{dd}_{[0,1],\mathbf{all}} = \mathfrak{dd}$ and, if $X' \subseteq X$ and $Y' \supseteq Y$, then

$$\mathfrak{dd}_{X',Y'} \leq \mathfrak{dd}_{X,Y}$$

Let's now provide the Galois-Tukey framework for $\mathfrak{dd}_{X,Y}$:

$$\mathfrak{d}(D_X, \mathrm{Sym}(\omega), R_Y) = \mathfrak{dd}_{X,Y}$$

where D_X is the family of infinite-coinfinite sets A with $d(A) \in X$ and $AR_Y p$ if $d(p[A]) \in Y$ and $d(p[A]) \neq d(A)$, for $A \in D_X$ and $p \in \mathrm{Sym}(\omega)$.

Let's then rephrase the main results of the last section in this framework:

Theorem 3.5. The following hold:

1. If $\mathbf{osc} \notin X$ and $\mathbf{osc} \in Y$, then

$$(D_X, \mathrm{Sym}(\omega), R_Y) \leq_T (\mathcal{M}, 2^\omega, \not\ni);$$

2. If $0 \in X$ or $1 \in X$, then

$$(\omega^\omega, \Phi_h, \in^\infty) \leq_T (D_X, \mathrm{Sym}(\omega), R_Y)$$

for $h \in \omega^\omega$ with $h(n) \geq 2^n + n + 1$, for all $n \geq 0$.

Proof. 1. Is easy to see that follows from Lemma 3.2 since, if $\mathbf{osc} \notin X$ and $\mathbf{osc} \in Y$, we can repeat the construction in the proof and find some $p \in \mathrm{Sym}(\omega)$ such that $d(p[A]) = \mathbf{osc} \in Y \setminus X$.

2. Eu não consigo ver como alterar a construção de forma que $d(\varphi_-(g)) = 1$ ao invés de 0. ⊣

This result already characterizes a lot of $\mathfrak{dd}_{X,Y}$ as $\mathbf{non}(\mathcal{M})$. We will see later that, in fact, this result can't be improved in the following sense: the assumptions in 1 and 2 are necessary.

With such characterization, it's natural to look for other bounds of $\mathfrak{dd}_{X,Y}$ for when $0, 1 \notin X$, $\mathbf{osc} \in X$ or $\mathbf{osc} \notin Y$. We will mostly obtain lower bounds, but there is one upper bound too.

Theorem 3.6. If $X \cap [0,1] \neq \emptyset$, then

$$(2^\omega, \mathcal{N}, \in) \leq_T (D_X, \mathrm{Sym}(\omega), R_Y)$$

Proof. The case when $0 \in X$ or $1 \in X$ can be reduced to the previous theorem if we could prove thar $(2^\omega, \mathcal{N}, \in) \cong (\omega^\omega, \mathcal{N}, \in) \leq_T (\omega^\omega, \Phi_h, \in^\infty)$, which is what we will gonna prove:

Consider the product measure $\mu = \prod_n \mu_n$ on ω^ω, where μ_n is a measure on ω defined as follows: fix a series $\sum_k a_k^n = 1$ such that $0 < a_k^n < \frac{1}{2^{n+1}h(n)}$, for all $k \in \omega$. Since n is fixed, we can easily find such series. Now, given $A \subseteq \omega$, let $\mu_n(A) = \sum_{k \in A} a_k^n$.

"With such measure on ω^ω we"...???

Let then $0, 1 \notin X$ and $r \in (0,1) \cap X$. Pendente ⊣

It's worth noting that it's an open question if the previous theorem holds if $X = \{\mathbf{osc}\}$.

Corollary 3.1. If $X \cap [0,1] \neq \emptyset$, then $\mathbf{cov}(\mathcal{N}) \leq \mathfrak{dd}_{X,Y}$

we can also compare $\mathfrak{dd}_{X,Y}$ with $\mathfrak{b}$ and $\mathfrak{d}$ when $\mathbf{osc} \in X$ or $\mathbf{osc} \notin Y$ as follows

Theorem 3.7. If $\mathbf{osc} \in X$ or $\mathbf{osc} \notin Y$, then

$$(\omega^\omega, \omega^\omega, \not\geq^*) \leq_T (D_X, \mathrm{Sym}(\omega), R_Y)$$

This proof is gigantic, so let's break it in some proof-sketches, one for each case.

Sketch. Assume first osc $\in X$, we want $\varphi_- : \omega^\omega \to D_X$ and $\varphi_+ : \mathrm{Sym}(\omega) \to \omega^\omega$ such that

$$\forall f \in \omega^\omega, \forall p \in \mathrm{Sym}(\omega),\ f \geq^* \varphi_+(p) \implies d(p[\varphi_-(f)]) = d(\varphi_-(f)) \in Y$$

Since we only know osc $\in X$, we want $\varphi_-(f)$ s.t. $d(\varphi_-(f)) = \text{osc}$. In fact, following the intuition of Lemma 3.6, we will construct such disjoint intervals using the growth of f, so we first need to choose $\varphi_+(p)$ in a way that $f \geq^* \varphi_+(p)$ guarantees us that we can let $a_n = f(a_{n-1})$ increases fast. Let, for now, $\varphi_+(p)(n) \geq 2^n$, then $f(n) \geq 2^n$ eventually, i.e., $\forall n \geq n_0$, let's then define $a_0^f = 0$ and

$$a_n^f = \begin{cases} 2^{a_{n-1}^f}, & \text{if } n < n_0 \\ f(a_{n-1}^f), & \text{if } n \geq n_0 \end{cases}$$

we could then just let $\varphi_-(f) = \bigcup_{n\geq 0}[a_{2n}^f, a_{2n+1}^f)$, and try to adjust $\varphi_+(p)$ in such a way that we can control where $p(k)$ goes knowing that $k < a_{2n}^f$ or $k > a_{2n+1}^f$ but, to do that in a satisfactory way, it's better to consider a middle point between a_{2n}^f and a_{2n+1}^f such that everyone from the left outside of this interval remains to the left, and similarly for the right.
Let then $\varphi_-(f) = \bigcup_{n\geq 0}[a_{4n}^f, a_{4n+2}^f)$. We want that if $n_0 < k < a_{4n}^f$, then $p(k) < a_{4n+1}^f = f(a_{4n+1}^f)$. Then, it would be good if we had $p(k) < \varphi_+(p)(k)$, since that could imply

$$p(k) < \varphi_+(p)(k) < \varphi_+(p)(a_{4n}^f) \leq f(a_{4n}^f) = a_{4n+1}^f$$

let then $\varphi_+(p)(n) \geq \max\{p(k) : k \leq n\}$. And, since we also want $p^{-1}(k) < a_{4n+2}^f$, whenever $k < a_{4n+1}^f$, we can then finally define

$$\varphi_+(p)(n) = \max\{p(k), p^{-1}(k) : k \leq n\} + 2^n$$

Proof. Assume osc $\in X$ and let $\varphi_+(p)(n) = \max\{p(k), p^{-1}(k) : k \leq n\} + 2^n$. Since there is some $n_0 \in \omega$ such that $f(n) \geq \varphi_+(p)(n)$, $\forall n \geq n_0$, let $a_0^f = 0$ and

$$a_n^f = \begin{cases} 2^{a_{n-1}^f}, & \text{if } n < n_0 \\ f(a_{n-1}^f), & \text{if } n \geq n_0 \end{cases}$$

then, define $\varphi_-(f) = \bigcup_{n\geq 0}[a^f_{4n}, a^f_{4n+2})$. So, as we saw, if $k < a^f_{4n}$, then $p(k) < a^f_{4n+1}$ and, if $k \geq a^f_{4n+2}$, then $p(k) \geq a^f_{4n+1}$; so $p[a^f_{4n+2}] = a^f_{4n+2}$. Now, we want to show that

$$|p[\varphi_-(f)] \cap a^f_{4n+1}| = |\varphi_-(f) \cap a^f_{4n+1}|$$

but it's not that hard to see. By the mere definition of $\varphi_-(f)$, the elements before a^f_{4n+1} are the ones until "half" of $I_n = [a^f_{4n}, a^f_{4n+2})$; and the ones that get mapped to elements before a^f_{4n+1} are, as we saw, first all the elements before I_n, and then $a^f_{4n+1} - a^f_{4n}$ elements inside I_n, resulting in the same amount. Similarly, using $p[a^f_{4n+2}] = a^f_{4n+2}$, we can conclude that

$$|p[\varphi_-(f)] \cap a^f_{4n+3}| = |\varphi_-(f) \cap a^f_{4n+3}|$$

but then, we have that

$$d_{a^f_{4n+1}}(p[\varphi_-(f)]) = d_{a^f_{4n+1}}(\varphi_-(f)) \geq \frac{a^f_{4n+1} - a^f_{4n}}{a^f_{4n+1}} = 1 - \frac{a^f_{4n}}{a^f_{4n+1}} \to 1$$

$$d_{a^f_{4n+3}}(p[\varphi_-(f)]) = d_{a^f_{4n+3}}(\varphi_-(f)) \leq \frac{a^f_{4n+2}}{a^f_{4n+3}} \to 0$$

so $d(p[\varphi_-(f)]) = \text{osc}$. ⊣

Sketch. Now, let's assume osc $\notin Y$. In order to illustrate the main idea, let's first consider the case where $\frac{1}{2} \in X$. Say $p \in \text{Sym}(\omega)$ is big on $E = 2\mathbb{N}$ if there is some $k = k_p$ such that there are infinitely many m satisfying

$$d_m(p[E]) > \frac{1}{2} + \frac{1}{k} \tag{6}$$

Similarly, it's small on E if there is some $k = k_p$ such that for infinitely many m's

$$d_m(p[E]) < \frac{1}{2} - \frac{1}{k} \tag{7}$$

The idea is, if our p is big (or small) on E, we can, as before, assume $f \geq^* \varphi_+(p)$ and choose it such that we can make a fast growing sequence $a_n = f(a_{n-1})$ and, with this, define

$$\varphi_-(f) = \bigcup_{n\in\omega}([a^f_{2n}, a^f_{2n+1}) \cap E) \cup \bigcup_{n\in\omega}([a^f_{2n+1}, a^f_{2n+2}) \cap (\omega \setminus E)) \tag{8}$$

then, intuitively, $d(\varphi_-(f)) = \frac{1}{2}$, but its rearrangement will oscillate, since we are inside E in a big chunk, but then outside of it, so p in $\varphi_-(f)$ will be sometimes small, sometimes big.

Since osc $\notin Y$, this would be enough. Lastly, for the case where p is neither big nor small on E, we just need to show that $p[\varphi_-(f)]$ has a subsequence of the density converging to $1/2$, then it's either converges to $1/2$ or oscillates. So, in both cases, we won't have $\varphi_-(f) R_Y p$.

Proof. If p is big (small) on E, let

$$m^p_n = \min\{m \geq 2^n : m \text{ satisfies eq. (6)(eq. (7))}\}$$

otherwise, if p is neither big nor small, let

$$m_n^p = \min\left\{m \geq 2^n : \frac{1}{2} - \frac{1}{2^n} \leq d_m(p[E]) \leq \frac{1}{2} + \frac{1}{2^n}\right\}$$

finally, let's define

$$\varphi_+(p)(n) = \min\{k > m_n^p : p^{-1}[m_n^p] \subseteq k\}$$

Now, if $f \geq^* \varphi_+(p)$, then there is some $n_0 \in \omega$ such that $f(n) \geq \varphi_+(p)(n) \geq m_n^p \geq 2^n$, $\forall n \geq n_0$. Now, define (a_n) as in the previous proof, and let $\varphi_-(f)$ as in (8).

First of all, let's prove properly that $d(\varphi_-(p)) = \frac{1}{2}$. Given some $m \geq n_0$, we know $m \in [a_k^f, a_{k+1}^f)$ for some $k \in \omega$. Since $|[a_i^f, a_{i+1}^f) \cap E| = \frac{a_{i+1}^f - a_i^f}{2} + \varepsilon_i$, where the error $\varepsilon_i \in \{0, \pm 1\}$ (similarly for O), then, we have that

$$\begin{aligned}
d_m(\varphi_-(p)) &= \frac{1}{m}\left(\sum_{n=0}^{k-1}\left(\frac{a_{n+1}^f - a_n^f}{2} + \varepsilon_n\right) + \left(\frac{m - a_k^f}{2} + \varepsilon_k\right)\right) \\
&= \frac{a_k^f - \cancelto{0}{a_0^f}}{2m} + \frac{m - a_k^f}{2m} + \frac{\varepsilon}{m} \\
&= \frac{1}{2} + \frac{\varepsilon}{m}
\end{aligned}$$

Since $\varepsilon \leq k$, and $m \geq a_k^f \geq 2^k$, then $\varepsilon/m \to 0$.

Now let's divide in two cases. First, assume p is big on E (small is similar) and fix $2n \geq n_0$, then

$$2^{a_{2n}^f} \leq m_{a_{2n}^f}^p < \varphi_+(p)(a_{2n}^f) \leq f(a_{2n}^f) = a_{2n+1}^f$$

We will now prove a seemly random inequality that will actually be important later:

Since $a_{2n+1}^f \geq \varphi_+(p)(a_{2n}^f)$, by definition we have $p^{-1}\left(m_{a_{2n}^f}^p\right) \subseteq a_{2n+1}^f$, then $m_{a_{2n}^f}^p \subseteq p[a_{2n+1}^f]$. In particular, every number greater than a_{2n+1}^f has image under p greater than $m_{a_{2n}^f}^p$. This implies that every number in $p[E] \cap m_{a_{2n}^f}^p$ is automatically in $p[E \cap a_{2n+1}^f] \cap m_{a_{2n}^f}^p$, so

$$d_{m_{a_{2n}^f}^p}(p[E \cap a_{2n+1}^f]) = d_{m_{a_{2n}^f}^p}(p[E]) > \frac{1}{2} + \frac{1}{k_p} \tag{9}$$

Now, with this in hands, we can calculate $d_{m_{a_{2n}^f}^p}(p[\varphi_-(f)])$ and $d_{m_{a_{2n+1}^f}^p}(p[\varphi_-(f)])$ and show it oscillates. Fix a_{2n+1}^f, then $\varphi_-(f)$ agrees with E from a_{2n}^f to a_{2n+1}^f, then $p[\varphi_-(f)]$ and $p[E]$ also, so

$$\left|p[\varphi_-(f)] \cap m_{a_{2n}^f}^p\right| \geq \left|p[\varphi_-(f) \cap a_{2n+1}^f] \cap m_{a_{2n}^f}^p\right| \geq \left|p[E \cap a_{2n+1}^f] \cap m_{a_{2n}^f}^p\right| - a_{2n}^f$$

i.e., they disagree by at most a_{2n}^f elements. So, by equation (9):

$$d_{m_{a_{2n}^f}^p}(p[\varphi_-(f)]) \geq d_{m_{a_{2n}^f}^p}(p[E]) - \frac{a_{2n}^f}{m_{a_{2n}^f}^p} > \frac{1}{2} + \frac{1}{k_p} - \frac{a_{2n}^f}{2^{a_{2n}^f}} > \frac{1}{2} + \frac{1}{k_p}$$

Similarly, by equation (9) we have for $p[O]$ a density $< \frac{1}{2} - \frac{1}{k_p}$, and the argument follows analogously. So we have $d_{m^p_{a^f_{2n}}}(p[\varphi_-(f)]) > \frac{1}{2} + \frac{1}{k_p}$ and $d_{m^p_{a^f_{2n+1}}}(p[\varphi_-(f)]) < \frac{1}{2} - \frac{1}{k_p}$, thus $d(p[\varphi_-(f)]) = \text{osc}$.

Let's now assume p is neither big nor small on E. The equality in equation (9) didn't assume p was big or small, so we know that

$$d_{m^p_{a^f_n}}(p[E \cap a^f_{n+1}]) \to \frac{1}{2}$$

in particular, the same is true for $\omega \setminus E$ instead. Then, within such subsequence, we can conclude that $d_{m^p_{a^f_n}}(p[\varphi_-(f)]) \to \frac{1}{2}$, which means its density is either $1/2$ or oscillates, as desired.

Let's now analyze the general case. If $0 \in X$ or $1 \in X$, the result follows by Theorem 3.5. Let's then assume $X \subseteq (0,1)$. Given $r \in (0,1)$, let:

and Cichoń's Diagram?

- $(\ell_b : b \in \omega)$ be a sequence such that $\left| r - \frac{\ell_b}{2^b} \right| < \frac{1}{2^b}$;
- $(j_b : b \in \omega)$ be such that $j_0 = 0$ and $j_{b+1} = j_b + 2^{2b}$ and $J_b = [j_b, j_{b+1})$;
- $J_b = I^0_b \mid I^1_b \mid \ldots \mid I^{2^b-1}_b$ be a uniform partition of J_b into 2^b intervals of length 2^b;
- for any $c \in \omega$, $[2^c]^{\ell_c} = \{e_c(v) : v < \omega_c\}$ be an enumeration of $[2^c]^{\ell_c}$, where $\omega_c = |[2^c]^{\ell_c}| = \binom{2^c}{\ell_c}$ and $e_c(0) = \ell_c$;
- for any $c \in \omega$ and $b \geq c$, $I^b_a = L^{a,c}_{b,0} \mid L^{a,c}_{b,1} \mid \ldots \mid L^{a,c}_{b,2^c-1}$ be a uniform partition of I^b_a into 2^c intervals of length 2^{b-c};
- for $v < \omega_c$, $E^c_v \subseteq \omega$ is such that $E^c_v \cap j_c = \emptyset$ and, for $b \geq c$ and $a \in 2^b$,

$$E^c_v \cap I^a_b = \bigcup_{d \in e_c(v)} L^{a,c}_{b,d}$$

Let's first show that E^c_v in fact has density $\ell_c/2^c$: First of all, note that from each I^a_b we will choose ℓ_c sets $L^{a,c}_{b,d}$, whose size is 2^{b-c}, then

$$|E^c_v \cap I^a_b| = \ell_c \cdot 2^{b-c} = \frac{\ell_c}{2^c} \cdot 2^b$$

which implies that

$$|E^c_v \cap J_b| = \frac{\ell_c}{2^c} \cdot |J_b| = \frac{\ell_c}{2^c} \cdot 2^{2b} \tag{10}$$

Now, let $n \in \omega$ be large enough, then $n \in J_b$ for some b, write $n = j_b + k \cdot 2^b + r_0$, where k is the number of I^b_a blocks before n, and r_0 the remainder for the incomplete block.

Now, from 0 to j_c, E^c_v has no elements. From j_c to j_b, by equation (10), we have $\frac{\ell_c}{2^c}(j_b - j_c)$ elements. From j_b to n we have $\frac{\ell_c}{2^c} \cdot (k \cdot 2^b)$ and a remainder of $\widetilde{r} \leq r_0$, therefore

$$d_n(E^c_v) = \frac{\ell_c}{2^c}\left(\frac{j_b - j_c + k \cdot 2^b}{n}\right) + \frac{\widetilde{r}}{n} = \frac{\ell_c}{2^c}\left(\frac{n - j_c - r_0}{n}\right) + \frac{\widetilde{r}}{n} = \frac{\ell_c}{2^c} - \frac{\ell_c}{2^c}\left(\frac{j_c + r_0}{n}\right) + \frac{\widetilde{r}}{n}$$

now, $\frac{\widetilde{r}}{n} \leq \frac{2^b}{j_b} \to 0$, and, similarly, $\frac{j_c + r_0}{n} \to 0$, so $d_n(E_v^c) \to \frac{\ell_c}{2^c}$.

Now, as in the previous case, we call $p \in \mathrm{Sym}(\omega)$ big if there is some $k = k_p$ such that for infinitely many c and m, we have $d_m(p[E_0^c]) > \frac{\ell_c}{2^c} + \frac{1}{k}$. Analogously for small and neither. We let m_n^p, $\varphi_+(p)$ and a_n^f, where $f \geq^* \varphi_+(p)$, as before.

We then define $(K_c : c \in \omega)$ an interval partition of ω into intervals of length ω_c. Now, if n is the v-th element of K_c, for $a_n^f \leq b < a_{n+1}^f$ and $a \in 2^b$, define

$$\varphi_-(f) \cap I_b^a = \bigcup_{d \in e_c(v)} L_{b,d}^{a,c}$$

Now, in order to show that $d(\varphi_-(f)) = r$, we follow a similar approach as E_v^c. Let $m \in \omega$, then $m \in J_b$ for some b. Also, for each J_b, with $b \in [a_n^f, a_{n+1}^f)$, we have a unique K_c and v associated, in particular, we can associate each J_b with some $E_{v(b)}^{c(b)}$ such that $\varphi_-(f) \cap [a_n^f, a_{n+1}^f) = E_{v(b)}^{c(b)} \cap [a_n^f, a_{n+1}^f)$.

Let then $m = j_b + k \cdot 2^b + r_0$ as before. Let's now use some asymptotic notation (because it's easier, since we already know the argument). Since $\frac{\ell_c}{2^c} \to r$, then from some $c(b_0)$ large enough, we have:

$$\begin{aligned}
|\varphi_-(f) \cap m| &= \sum_{i=0}^{b-1} |\varphi_-(f) \cap J_i| + k|\varphi_-(f) \cap I_b^a| + \widetilde{r} \\
&= \sum_{i=0}^{b-1} \frac{\ell_{c(i)}}{2^{c(i)}} 2^{2i} + k \frac{\ell_{c(b)}}{2^{c(b)}} 2^b + \widetilde{r} \\
&\sim C + \sum_{i=b_0}^{b-1} r 2^{2i} + kr 2^b + \widetilde{r} \\
&= C + r(m - j_{b_0} - r_0) + \widetilde{r}
\end{aligned}$$

then, dividing by m, we finally have that

$$d_m(\varphi_-(f)) \sim \cancelto{0}{\frac{C}{m}} + r\left(1 - \cancelto{0}{\frac{j_{b_0} - r_0}{m}}\right) + \cancelto{0}{\frac{\widetilde{r}}{m}} \to r$$

Let's now first note two important things:

1. We know by definition that $m = m^p_{a_n^f} < \varphi_+(p)(a_n^f) \leq f(a_n^f) = a_{n+1}^f$, then $p^{-1}[m] \subseteq a_{n+1}^f$, so

$$\begin{aligned}
|p[\varphi_-(f)] \cap m| &= |p[\varphi_-(f) \cap a_{n+1}^f] \cap m| \\
|p[E_v^c] \cap m| &= |p[E_v^c \cap a_{n+1}^f] \cap m|
\end{aligned}$$

   this implies that, since $\varphi_-(f) \cap [a_n^f, a_{n+1}^f) = E_v^c \cap [a_n^f, a_{n+1}^f)$ they differ only at $[0, a_n^f)$, so

$$\frac{|p[\varphi_-(f)] \cap m| - |p[E_v^c] \cap m|}{m} \leq \frac{a_n^f}{m} = \frac{a_n^f}{m^p_{a_n^f}} \to 0$$

   then $d_{m^p_{a_n^f}}(p[\varphi_-(f)]) \sim d_{m^p_{a_n^f}}(p[E_v^c])$, where E_v^c is chosen in function of n.

2. Since $d(E^c_v) = \frac{\ell_c}{2^c}$, if $d_m(E^c_0) > \frac{\ell_c}{2^c} + \frac{1}{k}$, then we can find another subset $e_c(v)$ instead of $e_c(0)$ such that its elements compensate and $d_m(E^c_v) < \frac{\ell_c}{2^c}$.

We can now break in cases:

Case 1. p **is big:** The case where p is small is analogous. In this case, let $n(c) \geq n_0$ be the 0-th item of K_c, then its corresponding set is E^c_0, so, by (1), we have that

$$d_{m^p_{a^f_{n(c)}}}(p[\varphi_-(f)]) \sim d_{m^p_{a^f_{n(c)}}}(p[E^c_0]) > \frac{\ell_c}{2^c} + \frac{1}{k_p}$$

but, by (2), if $n'(c)$ is the $v(c)$-th element of K_c, we have that

$$d_{m^p_{a^f_{n'(c)}}}(p[\varphi_-(f)]) \sim d_{m^p_{a^f_{n'(c)}}}(p[E^c_{v(c)}]) < \frac{\ell_c}{2^c}$$

so $d(\varphi_-(f)) = \text{osc}$, as desired.

Case 2. p **is neither big nor small:** then there isn't such k_p as in the definition, which means we can find a sequence $k_c \to \infty$ such that

$$\frac{\ell_c}{2^c} - \frac{1}{k_c} \leq d_{m^p_{a^f_{n(c)}}}(p[\varphi_-(f)]) \sim d_{m^p_{a^f_{n(c)}}}(p[E^c_0]) \leq \frac{\ell_c}{2^c} + \frac{1}{k_c}$$

which implies that, when $c \to \infty$, that $d_{m^p_{a^f_{n(c)}}}(p[\varphi_-(f)]) \to r$, so it either oscillates, or converges to r, as desired. $\dashv$

Corollary 3.2. If **osc** $\in X$ or **osc** $\notin Y$, then $\mathfrak{b} \leq \mathfrak{dd}_{X,Y}$

Pendente

Consider the simple relational system $(\mathfrak{c}, [\mathfrak{c}]^{\aleph_0}, \in)$, let's first prove some properties about it.

Lemma 3.4. $\mathfrak{d}(\mathfrak{c}, [\mathfrak{c}]^{\aleph_0}, \in) = \mathfrak{c}$ and $\mathfrak{b}(\mathfrak{c}, [\mathfrak{c}]^{\aleph_0}, \in) = \aleph_1$.

Proof. For the first equality, note that

$$\mathfrak{d}(\mathfrak{c}, [\mathfrak{c}]^{\aleph_0}, \in) \leq |[\mathfrak{c}]^{\aleph_0}| = \mathfrak{c}^{\aleph_0} = \left(2^{\aleph_0}\right)^{\aleph_0} = 2^{\aleph_0 \cdot \aleph_0} = 2^{\aleph}_0 = \mathfrak{c}$$

Now, given $A = \{A_\beta\}_{\beta<\kappa} \subseteq [\mathfrak{c}]^{\aleph_0}$ with $\kappa < \mathfrak{c}$, if $U = \bigcup A$, then

$$|U| \leq \sum_{\beta<\kappa} |A_i| \leq \sum_{\beta<\kappa} \aleph_0 = \kappa \cdot \aleph_0 < \mathfrak{c}$$

and, therefore, there is a $\alpha < \mathfrak{c}$ such that $\alpha \notin U$.

For the second equality, to show it's uncountable, let $X \subseteq \mathfrak{c}$ and just take $Y = X \in [\mathfrak{c}]^{\aleph_0}$, then, for every $\alpha \in X$, we have $\alpha \in Y$ too. Now, given any $X \subseteq \mathfrak{c}$ with $|X| = \aleph_1$, if $Y \in [\mathfrak{c}]^{\aleph_0}$, since it's countable, there is some $\alpha \in X$ such that $\alpha \notin Y$. $\dashv$

In order to prove the next theorem, we also need the following lemmas about almost disjoint families

Lemma 3.5. Let $A \subseteq \omega$ be any infinite set, then there is an almost disjoint family $\{A_\alpha : \alpha < \mathfrak{c}\}$ of subsets of A, where almost disjoint means that for any $\alpha \neq \beta < \mathfrak{c}$, we have that $|A_\alpha \cap A_\beta| < \aleph_0$.

Proof. Since $A \subseteq \omega$ is infinite, there is a bijection $f : \mathbb{Q} \to A$, then it's sufficient to find an almost disjoint family $\{A_\alpha : \alpha < \mathfrak{c}\}$ of subsets of $\mathbb{Q}$. Then, for each irrational number α, let $\{q_n^\alpha\}_{n\in\omega} \subseteq \mathbb{Q}$ be a sequence converging to α, and let $A_\alpha = \{q_n^\alpha : n \in \omega\}$. Then, since α is irrational, each A_α is infinite and, if $\alpha \neq \beta$, $q_n^\alpha = q_n^\beta$ only for finitely many n's, so $\{f(A_\alpha) : \alpha < \mathfrak{c}\}$ is our family. $\dashv$

This one is a little more technical and less general than the previous one, but we will use it to prove the next theorem

Lemma 3.6. There is an almost disjoint family $\{A_\alpha : \alpha < \mathfrak{c}\}$ of subsets of ω with density = osc.

Proof. Let $\{B_\alpha : \alpha < \mathfrak{c}\}$ be an almost disjoint family, guaranteed to exists by the previous lemma. Let's then define A_α, for each $\alpha < \mathfrak{c}$ as follows:

$$A_\alpha := \bigcup_{n\in B_\alpha} P_n, \text{ where } P_n = [(2n)!, (2n+1)!) \cap \omega$$

Clearly $\{P_n\}_{n\in\omega}$ are disjoint and, if $\alpha \neq \beta$, then, by definition, $B_\alpha \cap B_\beta = \{n_1, n_2, \ldots, n_m\}$ is finite, and $A_\alpha \cap A_\beta = \cup_{k=1}^m P_k$, which is also finite. Now, let $B_\alpha = \{b_0, b_1, \ldots\}$, then, for $N = (2b_n + 1)!$, we have $P_{b_n} \subseteq A_\alpha \cap N$, which implies that

$$d_N(A_\alpha) = \frac{|A_\alpha \cap N|}{N} \geq \frac{(2b_n+1)! - (2b_n)!}{(2b_n+1)!} = 1 - \frac{(2b_n)!}{(2b_n+1)!} \to 1$$

However, if we let $N = (2b_n)!$, then we have that

$$d_N(A_\alpha) = \frac{|A_\alpha \cap N|}{N} \leq \frac{(2b_n - 1)!}{(2b_n)!} \to 0$$

So $d(A_\alpha) = \text{osc}$, as desired. $\dashv$

Theorem 3.8. If ($0 \in X$ or $1 \in X$) and $\text{osc} \notin Y$, or if $\text{osc} \in X$ and ($0 \notin Y$ or $1 \notin Y$), then

$$(\mathfrak{c}, [\mathfrak{c}]^{\aleph_0}, \in) \leq_T (D_X, \text{Sym}(\omega), R_Y)$$

Sketch. We need a $\varphi_- : \mathfrak{c} \to D_X$, $\alpha \mapsto A_\alpha$ and a $\varphi_+ : \text{Sym}(\omega) \to [\mathfrak{c}]^{\aleph_0}$ such that

$$\forall \alpha < \mathfrak{c}, \forall p \in \text{Sym}(\omega),\ d(A_\alpha) \neq d(p[A_\alpha]) \in Y \Rightarrow \alpha \in \varphi_+(p)$$

It would be great if we could take $\varphi_+(p) = \{\alpha : d(A_\alpha) \neq d(p[A_\alpha]) \in Y\}$, but we don't know if such $\varphi_+(p)$ is in $[\mathfrak{c}]^{\aleph_0}$. Since we don't even know what is in Y, let's first consider the case where $0 \in X$, then it's sufficient to take $\varphi_+(p) = \{\alpha : d(p[A_\alpha]) > 0\}$, whenever $\mathrm{osc} \notin Y$. Now, how many sets there are in $\varphi_+(p)$? Well, we can write it as

$$\varphi_+(p) = \bigcup_{k\geq 1} \underbrace{\left\{\alpha : d(p[A_\alpha]) \geq \frac{1}{k}\right\}}_{B_k}$$

And, if we could guarantee that $d(p[A_\alpha \cup A_\beta]) = d(p[A_\alpha]) + d(p[A_\beta])$, then

$$\sum_{\alpha \in B_k} d(p[A_\alpha]) \geq \frac{|B_k|}{k}$$

but $|B_k|/k \leq 1$, otherwise $p[\bigcup_{\alpha\in B_k} A_\alpha]$ would have density > 1, then each $|B_k| \leq k$ is finite.

The easiest scenario would then be where A_α are all pairwsise disjoint, but can we write any infinite $A \subseteq \omega$ as a disjoint family of $\mathfrak{c}$ subsets of it? Of course you can't, otherwise you could select a different element from each subset, and then forming a subset of A with $\mathfrak{c}$ many elements.

This is where our previous lemma comes in, if A and B are almost disjoint, since density is blind for finitely many sets, then $d(A \cup B) = d(A) + d(B)$.

Proof. For the first case, assume that $0 \in X$ and $\mathrm{osc} \notin Y$. Let A be such that $d(A) = 0$, and let $\{\varphi_-(\alpha) = A_\alpha : \alpha < \mathfrak{c}\}$ be an almost disjoint family of subsets of A. Then $d(A_\alpha) = 0$ for all $\alpha < \mathfrak{c}$ and we can define $\varphi_-(\alpha) = A_\alpha$. Now, given $p \in \mathrm{Sym}(\omega)$, let $\varphi_+(p) = \{\alpha < \mathfrak{c} : d(p[A_\alpha]) > 0\}$, as we saw this is at most countable, so if it's finite we can just fill in with a bunch of other random $\beta < \mathfrak{c}$ to make it countable. The case where $1 \in X$ just consider $\varphi_-(\alpha) = \omega \setminus A_\alpha$, then $d(p[\omega \setminus A_\alpha]) = 1 - d(p[A_\alpha])$ and $\varphi_+(p) = \{\alpha < \mathfrak{c} : d(p[\varphi_-(\alpha)]) < 1\} = \{\alpha < \mathfrak{c} : d(p[A_\alpha]) > 0\}$.

For the second case, let $0 \notin Y$ (when $1 \notin Y$ just take complements as in $1 \in X$). Let, by lemma 3.6, $\{\varphi_-(\alpha) : \alpha < \mathfrak{c}\}$ be an almost disjoint family of subsets of ω without density. Then

$$\varphi_+(p) = \{\alpha < \mathfrak{c} : d(p[\varphi_-(\alpha)]) > 0\}$$

is at most countable and satisfies what we want. $\dashv$

Corollary 3.3. Under the assumptions of the previous theorem

$$\mathfrak{dd}_{X,Y} = \mathfrak{c} \text{ and } \mathfrak{dd}^\perp_{X,Y} \leq \aleph_1$$

In fact, we can't prove $\mathfrak{dd}^\perp_{X,Y} = \aleph_1$. For example, let $X = \{0\}$ and $Y = (\frac{1}{2}, 1]$, then given $F = \{A, B\}$ two disjoint infinite sets of density 0, if $d(p[A]) \in Y$, i.e., $d(p[A]) > \frac{1}{2}$, then $d(p[B]) \leq \frac{1}{2}$, so B witnesses $\neg BR_Y p$ and, therefore, $\mathfrak{dd}^\perp_{\{0\},(1/2,1]} = 2$. Analogously, let $Y_n = (\frac{1}{n}, 1]$, then, if $\{A_k : k \leq n\}$ is any disjoint family of infinite subsets of density 0, then, we can't have $d(p[A_k]) > \frac{1}{n}$ for all $k \leq n$, otherwise $p[A_1 \cup \cdots \cup A_n]$ would have density > 1. Then $\mathfrak{dd}^\perp_{\{0\},Y_n} = n$.

Let UR be the unreaping relation on $[\omega]^\omega$, where, $A\ UR\ B$ iff $B \subseteq^* A$ or $B \cap A$ is finite (i.e., A does not split B in infinitely many sets inside and outside A). $\mathfrak{d}([\omega]^\omega, [\omega]^\omega, UR) = \mathfrak{r}$ is called the reaping number, and $\mathfrak{b}([\omega]^\omega, [\omega]^\omega, UR) = \mathfrak{s}$ is called the splitting number.

Theorem 3.9. Assume $0, 1 \notin X$ and either

- $0, 1 \in Y$ or
- $\mathrm{osc} \notin X$ and $\exists r \in (0, 1]$ s.t. $\forall x \in X$ both $r + x(1 - r)$ and $x(1 - r)$ are in Y.

Then $(D_X, \mathrm{Sym}(\omega), R_Y) \leq_T ([\omega]^\omega, [\omega]^\omega, UR)$.

Pendente

Sketch. We need $\varphi_- : D_X \to [\omega]^\omega$ and $\varphi_+ : [\omega]^\omega \to \mathrm{Sym}(\omega)$ such that

$$y\ UR\ \varphi_-(x) \Rightarrow \varphi_+(y) R_Y x$$

Let's start with the first condition, where $0, 1 \notin X$ and $0, 1 \in Y$. The easiest φ_- we could take is the identity, then we need for each $y \in [\omega]^\omega$ a $p_y \in \mathrm{Sym}(\omega)$ such that

$$x \setminus y \text{ or } x \cap y \text{ is finite} \Rightarrow d(x) \neq d(p_y[x]) \in Y$$

In fact, since we only know $0, 1 \in Y$, we want p_y such that $d(p_y[x]) = 0$ or 1, and the fact that $0, 1 \notin X$ will already guarantee that $d(x) \neq d(p_y[x])$. Now, $x \setminus y$ being finite, with $x, y \in [\omega]^\omega$, basically says that y is almost inside x, except for finitely many elements. And $x \cap y$ being finite says y is almost outside x, except for finitely many elements. In both cases, since that finite amount does not affect the density in the limit, then we can see as if $y \subseteq x$ or $y \cap x = \emptyset$.

In particular, this shows we just need $a, 1 - a \notin X$ and $a, 1 - a \in Y$ for any $a \in [0, 1]$.

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Proof.

- Let $0, 1 \in Y$, $\varphi_-(x) = x$ and $\varphi_+(y) = p_y \in \mathrm{Sym}(\omega)$ be such that $d(p_y[y]) = 1$ for all $y \in [\omega]^\omega$. If $y \subseteq^* x$, then $F = y \setminus x$ is finite, and $y \subseteq x \cup F$, so

$$d_n(p_y[x] \cup p_y[F]) = \frac{|p_y[x] \cap n| + |p_y[F] \cap n|}{n} \geq \frac{|p_y[y] \cap n|}{n} = d_n(p_y[y])$$

  and, since $p_y[F]$ is finite, then the LHS converges to $d(p_y[x])$ and the RHS to $d(p_y[y]) = 1$.

  If $F = x \cap y = x \setminus (\omega \setminus y)$ is finite, then, by the previous argument, $d(p_y[x]) \to d(p_y[\omega \setminus y]) = 0$.

  Now, since $0, 1 \notin X$, then $d(x) \neq 0, 1$ and, therefore, $d(x) \neq d(p_y[x])$.

- Let $0, 1, \mathrm{osc} \notin X$, i.e., $X \subseteq (0, 1)$, and assume we have such $r \in (0, 1]$. Again, let $\varphi_-(x) = x$ and fix $z \subseteq \omega$ with $d(z) = r$. For $y \in [\omega]^\omega$, let $y' \subseteq y$ be s.t. $d(y') = 0$ and let $\varphi_+(y) = p_y$ be the unique permutation mapping y' to z and $\omega \setminus y'$ to $\omega \setminus z$, both in an order-preserving way.

  Now, since $x = (x \cap y') \sqcup (x \setminus y')$, then

$$d(p_y[x]) = d(p_y[x \cap y']) + d(p_y[x \setminus y']) \tag{11}$$

Let $\omega \setminus y' = \{a_0 < a_1 < \dots\}$ and $\omega \setminus z = \{b_0 < b_1 < \dots\}$, then $p_y(a_i) = b_i$ and, since $x \setminus y' \subseteq \omega \setminus y'$, we will have that, if $(\omega \setminus y') \cap n$ has k elements $a_0, a_1, \dots, a_{k-1}$, then

$$d^n_{\omega\setminus y'}(x \setminus y') = \frac{|\{i < k : a_i \in x \setminus y'\}|}{k}$$

and, as $n \to \infty$, $k \to \infty$, so $d^n_{\omega\setminus y'}(x \setminus y') \to d(x \setminus y')$, but $x = (x \setminus y') \sqcup (y' \cap x)$, where $y' \cap x \subseteq y'$, then $d(y' \cap x) = 0$ and, therefore, $d_{\omega\setminus y'}(x \setminus y') = d(x)$.

Now, since $p_y[\omega \setminus y'] = \omega \setminus z$, then, if $(\omega \setminus y') \cap n$ has k elements, then also has $(\omega \setminus z) \cap n$ and, by the same argument as before, we then have that

$$d^n_{\omega\setminus z}(p_y[x \setminus y']) = \frac{|i < k : b_i \in p_y[x \setminus y']|}{k} = \frac{|\{i < k : a_i \in x \setminus y'\}|}{k} \to d(x)$$

and, finally, we can then calculate one of the terms in (6), i.e.

$$d_n(p_y[x \setminus y']) = \frac{|p_y[x \setminus y'] \cap n|}{n} = \frac{|p_y[x \setminus y'] \cap n|}{|(\omega \setminus z) \cap n|} \cdot \frac{|(\omega \setminus z) \cap n|}{n} \to d(x)(1 - r)$$

Let's now, again, split in both cases: If $y \subseteq^* x$, then $y' \subseteq^* x$, $y' = (x \cap y') \sqcup (y' \setminus x)$ and, since $p_y[y'] = z$ and $y' \setminus x$ is finite, then also is $p_y[y' \setminus x]$, and $d(p_y[y' \setminus x]) = 0$, therefore

$$r = d(z) = d(p_y[x \cap y']) + d(p_y[y' \setminus x]) = d(p_y[x \cap y'])$$

So, from (6), $d(p_y[x]) = d(x)(1 - r) + r > d(x)$, since $1 \notin X$.

Now, if $x \cap y$ is finite, then also is $x \cap y'$, therefore $d(p_y[x]) = d(x)(1 - r) < d(x)$, since $0 \notin X$. $\dashv$

Corollary 3.4. Under the assumptions of the previous theorem

$$\mathfrak{dd}_{X,Y} \leq \mathfrak{r} \text{ and } \mathfrak{s} \leq \mathfrak{dd}^{\perp}_{X,Y}$$

Pendente

Theorem 3.10. For any $X \subseteq [0, 1]$, we have that

$$\mathfrak{dd}_{X,\{\mathrm{osc}\}} = \mathfrak{dd}_{X,\mathrm{all}}$$

Proof. Since $\{\mathrm{osc}\} \subseteq \mathrm{all}$, then $\mathfrak{dd}_{X,\mathrm{all}} \leq \mathfrak{dd}_{X,\{\mathrm{osc}\}}$, so we just need to show that $\mathfrak{dd}_{X,\{\mathrm{osc}\}} \leq \mathfrak{dd}_{X,\mathrm{all}}$.

Let $C \subseteq \mathrm{Sym}(\omega)$ be a witness for $\mathfrak{dd}_{X,\mathrm{all}}$, and let $\widetilde{C} = C \cup \{g_p : p \in C\}$, where g_p is the function that alternates between id and p in The Mix Lemma. Now, given $A \in D_X$, let $p \in C$ be such that $d(p[A]) \neq d(A)$. If $d(p[A]) = \mathrm{osc}$, we are done. Otherwise, we have $d(p[A]) \in [0, 1]$ and, therefore, $d(g_p[A]) = \mathrm{osc}$, which concludes the proof. $\dashv$

4 Sum and Density Preserving Permutations

Let's start this section proving a really famous theorem due to Agnew, which characterizes all sum preserving permutations

Definition 4.1. A permutation $p \in \mathrm{Sym}(\omega)$ is called Agnew if there exists a $k \in \omega$ such that for all $n \in \omega$, $p[n]$ can be written as a union of at most k intervals.

Equivalently, we can say that p is Agnew if we can switch between elements in $\mathrm{Im}(p)$ and outside at most k times. To formalize such condition, say $M < N$ when $\max(M) < \min(N)$, and call $M = \{m_0 < m_1 < \cdots < m_k\}$ and $N = \{n_0 < n_1 < \cdots < n_k\}$ collated sets if their elements alternate between each other, i.e., $m_0 < n_0 < m_1 < \cdots < m_k < n_k$.

Definition 4.2. We say $p \in \mathrm{Sym}(\omega)$ satisfies condition A if there exists a $k \in \omega$ such that whenever M and N are collated sets and $p[M] < p[N]$, we have $|M| = |N| < k$.

It's easy to see that p is Agnew iff p^{-1} satisfies condition A. We can now state one of the main Theorems of this section.

Theorem 4.1. $p \in \mathrm{Sym}(\omega)$ is Agnew iff it's a sum preserving permutation.

Sketch. ($\Rightarrow$) Given any $s = \sum_n a_n$, we know by Cauchy's Criterion that, varying m, the partial sums starting from a_m can be made arbitrarily small. So we can choose n such that $p[n]$ covers $[0, m]$, meaning all at most k gaps between the intervals in the image of p can be made arbitrarily small.

($\Leftarrow$) Assume p is not Agnew, we will find a $\sum_n a_n$ whose rearrangement corrupts. Since p is not Agnew, we can find arbitrarily large alternating sequences of elements in $\mathrm{Im}(p)$ and outside of it.

Fix two such elements relative to $p[n_1]$ and associate the one inside $\mathrm{Im}(p)$ with 1, and the other with -1. Now, let n_2 such that $p[n_2]$ covers the previous interval and guarantees us to have four new alternating elements. Associate the ones inside $\mathrm{Im}(p)$ with $\frac{1}{2}$, and the ones outside with $-\frac{1}{2}$. In general, choose n_k such that $p[n_k]$ covers all previous elements and guarantees us $2k$ new elements, half inside and half outside of $\mathrm{Im}(p)$. Associate the ones outside with $\frac{1}{k}$, and the ones inside with $-\frac{1}{k}$. Now, for all the remaining indices, just let them to be 0, this defines a sequence a_n, where a_n is the value we associate to its index n. Clearly for any k, we can let $\sum a_n$ be such that all its partial sums eventually has value at most $\frac{1}{k}$, meaning it converges to 0. However, by construction, the partial sums of the rearranged series up to n_k is always 1.

Let's now write all these steps more formally

Proof. ($\Rightarrow$) By Cauchy Criterion, given $\varepsilon > 0$, there is an $m_0 \in \omega$ such that $\forall n > m \geq m_0$, we have

$$\left|\sum_{i=m}^{n} a_i\right| < \frac{\varepsilon}{2k}$$

Let then n be large enough such that $p[n]$ covers $[0, m_0]$, then, since p is Agnew, we can write $p[n] = I_1 \cup I_2 \cup \cdots \cup I_\ell$, where $\ell \leq k$ and $I_i = [b_i, B_i]$. Therefore, since any $m \in I_j$ has $m \geq m_0$, for $j = 1, \ldots, \ell$, then, by Cauchy's Criterion

$$\left|\sum_{i=0}^{n} a_{p(i)} - \sum_{i=0}^{b_1} a_i\right| \leq \left|\sum_{i \in I_1} a_i\right| + \cdots + \left|\sum_{i \in I_\ell} a_i\right| < \frac{\varepsilon}{2}$$

Therefore, we have that

$$\left|\sum_{i=0}^{n} a_{p(i)} - \sum_{i=0}^{n} a_i\right| \leq \left|\sum_{i=0}^{n} a_{p(i)} - \sum_{i=0}^{b_1} a_i\right| + \left|\sum_{i=0}^{b_1} a_i - \sum_{i=0}^{n} a_i\right| < \varepsilon$$

($\Leftarrow$) É bem mais fácil de ver com um desenho colorido :) $\dashv$

It would be really nice if condition A also characterizes density preserving permutations. Unfortunately, this is not the case, we will see later that condition A preserves what we call strong density, or uniform density.

Now, as expected, the previous condition would not be called condition A if we didn't have a condition B:

Definition 4.3. We say $p \in \mathrm{Sym}(\omega)$ satisfies condition B if there exists a $k \in \omega$ such that whenever $|M| = |N|$, with $M < N$ and $p[N] < p[M]$, we have $|M| = |N| < k$.

It's not hard to see that condition B implies condition A, but the converse is not true. The following lemma actually shows something stronger:

Lemma 4.1. Condition B implies condition A, but we can find some p such that both p and p^{-1} satisfies A, but not B.

Proof. Consider M, N collated sets, i.e., $m_1 < n_1 < m_2 < n_2 < \cdots < m_j < n_j$, such that $p[M] < p[N]$, then, if ℓ is like a half of j (you could take the floor), then

$$n_1 < n_2 < \cdots < n_\ell < m_{\ell+1} < \cdots < m_j$$

or $j-1$ so both sets $N' = \{n_1, \ldots, n_\ell\}$ and $M' = \{m_{\ell+1}, \ldots, m_j\}$ have the same size. Therefore, $N' < M'$, but $p[M'] < p[N']$, then, by condition B, $|N'| = \ell < k$, meaning j is at most $2k$ (or $2k+1$ depending on the parity).

To construct the counterexample, let $J_1 = \{1\}$, $J_2 = \{2, 3\}$, $J_3 = \{4, 5, 6\}$, etc. And consider the unique p that biject J_k in J_k reserving the order of its elements. Then, for any k, we can take M as the first half of J_{2k} and N as the remainder, then $M < N$, but $p[N] < p[M]$. However, note that $p = p^{-1}$ and p is clearly Agnew, since the image is the union of at most 2 intervals. $\dashv$

It would then be expected, based on the discussions until now, that condition B would characterize the permutations that preserve relative density. However, this is also not the case:

Theorem 4.2. $p \in \mathrm{Sym}(\omega)$ satisfies condition B iff it preserves relative density.

Proof. ($\Rightarrow$) For any n, let $M = \{m < n : p(m) \geq n\}$ then, since p is a permutation, we can find some $N \subseteq \{m \geq n : p(m) < n\}$ with the same size as M. Then $M < N$, but $p[N] < p[M]$, so $|M| = |N| < k$, which implies that, if $A \subseteq \omega$ is infinte coinfinite with density, then

$$||p[A] \cap n| - |A \cap n|| < k \implies |p[A] \cap n| = |A \cap n| + m_n^A$$

with $|m_n^A| < k$. This shows p preserves density. Now, given $A \subseteq B \subseteq \omega$ infinite and coinfinite with relative density, by the previous result we have that

$$\frac{|p[A] \cap n|}{|p[B] \cap n|} = \frac{|A \cap n| + m_n^A}{|B \cap n| + m_n^B} = \frac{\frac{|A \cap n|}{|B \cap n|} + \frac{m_n^A}{|B \cap n|}}{1 + \frac{m_n^B}{|B \cap n|}} \to d_B(A)$$

then $d_{p[B]}(p[A]) = d_B(A)$, i.e., it preserves relative density.

($\Leftarrow$) This one is trickier. ⊣

What does such result reveal to us? Not that much, to be fair. Actually, it just says that the set of permutations that preserve relative density are also sum-preserving, but it says nothing about the set of permutations that preserve density, which is what we want.

Before we proceed, look again at the ($\Rightarrow$) case in the previous proof, we showed that condition B implies the following condition:

$$\lim_{n \to \infty} \frac{|\{m \in \omega : m < n \leq p(m)\}|}{n} = 0 \tag{L}$$

such condition L (due to Lévy) has some interesting properties that was deeply studied.

Definition 4.4. The set $\mathcal{G}$ of permutations that satisfies L form a group, known as Lévy's Group.

Proof. Pendente ⊣

And, in relation to what it preserves, we have the following theorem:

Theorem 4.3. $p \in \mathcal{G}$ if, and only if, for all $A \subseteq \omega$

$$\lim_{n \to \infty} d_n(A) - d_n(p[A]) = 0$$

Proof. ($\Rightarrow$) is a scholium of the start of the previous proof;

($\Leftarrow$) if $p \notin \mathcal{G}$, choose (a_n) fast growing such that $\frac{a_{n-1}}{a_n} \to 0$ and $E_n = \{m \in \omega : m < a_n \leq p(m)\}$ satisfies $\frac{|E_n|}{a_n} > \alpha$, for some $\alpha > 0$. Then, let $A = \bigcup_n E_n$. We have that

$$d_{a_n}(A) \geq d_{a_n}(E_n) > \alpha > 0$$

and, since $p[E_j] \geq n_j$, we have $p[A] \cap a_n \subseteq \bigcup_{j<n} p[E_j]$, so

$$d_{a_n}(p[A]) \leq d_{a_n}\left(p\left[\bigcup_{j<n} E_j\right]\right) = d_{a_n}\left(\bigcup_{j<n} E_j\right) \leq \frac{a_{n-1}}{a_n} \to 0$$

then $\lim_{n\to\infty} d_{a_n}(A) - d_{a_n}(p[A]) \geq \alpha > 0$. $\dashv$

i.e., the permutations that satisfies L "preserves" density in all cases, including when $d(A) = \text{osc}$. This shows that, if we had a condition C such that p satisfies C iff p preserves density, then condition L would imply it. Let's show such implications are strict.

Theorem 4.4. $\mathrm{B} > \mathrm{L} > \mathrm{C}$ and $\mathrm{A} \perp \mathrm{L}, \mathrm{C}$.

Proof. We already know that $\mathrm{B} \geq \mathrm{L} \geq \mathrm{C}$, then it just remains to show the strict inequalities and the incomparability with condition A.

- $\mathrm{A} \not\geq \mathrm{L}$: Let $N_{k+1} \gg N_k$, and consider p that swaps $[N_k, 2N_k)$ with $[2N_k, 3N_k)$, for each $k \in \omega$. Clearly p satisfies condition A, since $p^{-1} = p$ is Agnew (the image $p[n]$ is always the union of at most two intervals). However, using the notation of the previous proof, we have $|E_{2N_k}| = N_k$, then it doesn't satisfy condition L.

- $\mathrm{C} \not\geq \mathrm{L}$: The permutation defined in the previous item works, since

$$\begin{aligned} |p[A] \cap n| &= |\{k < n : p(k) < n\}| + |\{k \geq n : p(k) < n\}| \\ &= |A \cap n| - |\{k < n : p(k) \geq n\}| + |\{k \geq n : p(k) < n\}| \end{aligned}$$

- $\mathrm{L} \not\geq \mathrm{B}$: We now need to swap two chunks, but whose size grows sub linearly in some sense. Let then $M_k = [2^k, 2^k + k)$ and $N_k = [2^k + k, 2^k + 2k)$ and p be the unique permutation that swap both, for each $k \in \omega$. Then, clearly p doesn't satisfy condition B, however, for $2^k \leq n < 2^{k+1}$, the maximum amount E_n can attain is at $n = 2^k + k$, where

$$\frac{|E_n|}{n} = \frac{k}{2^k + k} \to 0$$

  so p does satisfy condition L.

  It's worth noting that this proof was unnecessary: if $\mathrm{L} \geq \mathrm{B}$, since $\mathrm{B} \geq \mathrm{A}$, we would have $\mathrm{L} \geq \mathrm{A}$, contradicting the next item.

- $\mathrm{L} \not\geq \mathrm{A}$: The heuristic is clear, based on the previous one. We just let $J_k = [2^k, 2^k + 2k)$, and let p be such that the even numbers in J_k are mapped to its first k elements, and the odd ones to the last k. Then, by the same argument as before, p satisfies condition L, but we have arbitrarily large collated sets that get separated, so it doesn't satify condition A.

- $\mathrm{C} \not\geq \mathrm{A}$: Since $\mathrm{L} \geq \mathrm{C}$, if $\mathrm{C} \geq \mathrm{A}$, then $\mathrm{L} \geq \mathrm{A}$, which is false by the previous item.

- A $\not\geq$ C: Let $a_0 = 0$ and $a_{k+1} = 2^{a_k}$, let p be such that it swaps the interval $I_k = [a_k, 2a_k)$ with $J_k = [a_k^2, a_k^2 + a_k)$, and let $A = 2\mathbb{N} \setminus \bigcup_k J_k$. Then, when $n \in J_k$, we have that $A \cap n$ has all the even elements until n, but at most the a_k ones in I_k. But, since $n \geq a_k^2$, at this point these missed elements are negligible, and then $d(A) = \frac{1}{2}$.

  However, for $n = 2a_k$, we have $p[A] \cap n$ missing all a_k elements in I_k, then, since $n = 2a_k$ and $p[A]$ are the even naturals with gaps, it has at most $\frac{a_k}{2}$ elements in it, i.e., $d_n(p[A]) \leq \frac{1}{4}$.

$\dashv$